Kent-AI: Conversational Case-Taking and Semantic Repertorization

A project report submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY



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Kent-AI: Conversational Case-Taking and Semantic Repertorization

ABSTRACT

Classical homeopathic prescribing requires the physician to match a patient's complete symptom totality — comprising mental, emotional, physical sensation, anatomical location, modality, concomitant, and temporal dimensions — against over 74,000 hierarchical rubrics indexed in Dr. James Tyler Kent's Repertory. This process is cognitively demanding, time-consuming, and subject to practitioner recall bias. Kent-AI is an AI-powered conversational clinical decision support system designed to automate and augment this workflow using state-of-the-art Natural Language Processing, dense semantic retrieval, and large language model reasoning.

The system integrates four fine-tuned biomedical transformer encoders (BioLinkBERT-Base, PubMedBERT, Bio_ClinicalBERT, and DeBERTa-v3) totalling 414 million parameters, trained on a bespoke synthetic dataset of 19,732 clinical cases annotated with 99,854 entities across 15 BIO token labels spanning 7 Kentian symptom dimensions. A ChromaDB HNSW vector index encodes all 74,513 rubrics using `all-MiniLM-L6-v2` sentence embeddings, enabling sub-18 millisecond semantic retrieval. A QLoRA-fine-tuned BioMistral 7B large language model serves as a clinical reasoning arbiter in a two-stage hybrid extraction pipeline, achieving 47.59% Micro F1, 100% JSON schema validity, and 99.73% clinical grounding on the held-out test set.

A doctor-in-the-loop Streamlit clinical workstation implements a 4-step consultation stepper (Intake, Review, Analysis, Report), an interactive Totality Matrix, a persistent Case Library, and a multi-page ReportLab PDF exporter. The system is validated by 195 automated tests across 22 test modules. BioLinkBERT achieves the highest Token-Level BIO F1 of 34.86%, PubMedBERT achieves the highest Dense Rubric Recall@20 of 41.56%, and DeBERTa-v3 achieves the highest Top-5 Simillimum Remedy Accuracy of 42.16% under

narrative-embedding mode. Kent-AI represents the first systematic application of large-scale biomedical NLP ensembling to classical homeopathic repertorization, with a fully open-source, clinically verifiable implementation.

Keywords: Homeopathic Repertorization, Biomedical Named Entity Recognition, Dense Retrieval, Clinical Decision Support, Large Language Models, Parameter-Efficient Fine-Tuning, QLoRA, ChromaDB.

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We are grateful to the **Department of Information Technology, IIIT Allahabad** for providing access to the high-performance computing infrastructure — specifically the **NVIDIA RTX 5000 Ada Generation GPU (32GB VRAM)** — which made the large-scale model fine-tuning experiments possible. Without this hardware access, training four biomedical transformers and two large language models simultaneously would not have been feasible within the project timeline.

We acknowledge the open-source community whose foundations we build upon: the HuggingFace Transformers and PEFT libraries, ChromaDB, Ollama, Streamlit, and the ReportLab PDF toolkit. Special recognition goes to the creators of Kent's digitized Repertory dataset which serves as the foundational knowledge base of this entire system.

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List of Abbreviations

AI	**A**rtificial **I**ntelligence
NLP	**N**atural **L**anguage **P**rocessing
NER	**N**amed **E**ntity **R**ecognition
BIO	**B**eginning-**I**nside-**O**utside (tagging scheme)
LLM	**L**arge **L**anguage **M**odel
PEFT	**P**arameter-**E**fficient **F**ine-**T**uning
QLoRA	**Q**uantized **Lo**w-**R**ank **A**daptation
HNSW	**H**ierarchical **N**avigable **S**mall **W**orld
FSM	**F**inite **S**tate **M**achine
MIND	Kent's Repertory Chapter 1: Mental and Emotional Symptoms
MENT	**Ment**al / Emotional Dimension (BIO label)
LOC	**Loc**ation Dimension (BIO label)
SEN	**Sen**sation Dimension (BIO label)
MOD_AGG	**Mod**ality **Agg**ravation Dimension (worse from)
MOD_AMEL	**Mod**ality **Amel**ioration Dimension (better from)
CONC	**Conc**omitant Symptoms Dimension
TEMP	**Temp**oral / Time Pattern Dimension
F1	F1-Score (Harmonic Mean of Precision and Recall)
MRR	**M**ean **R**eciprocal **R**ank
ChromaDB	Open-source vector database with HNSW cosine indexing
VRAM	**V**ideo **R**andom **A**ccess **M**emory
IIITA	**I**ndian **I**nstitute of **I**nformation **T**echnology, **A**llahabad
UI	**U**ser **I**nterface
UX	**U**ser e**X**perience
PDF	**P**ortable **D**ocument **F**ormat
API	**A**pplication **P**rogramming **I**nterface
REST	**Re**presentational **S**tate **T**ransfer
CUDA	**C**ompute **U**nified **D**evice **A**rchitecture (NVIDIA GPU Framework)
RTD	**R**eplaced **T**oken **D**etection (pre-training objective)
MIMIC	**M**edical **I**nformation **M**art for **I**ntensive **C**are

To the practitioners of classical homeopathy worldwide who dedicate their lives to the art and science of healing, and to the patients who place their trust in them.

Chapter 1

Introduction

This chapter provides an introduction to the Kent-AI project, a conversational clinical decision support system for homeopathic repertorization. It presents the clinical and technical motivation for the system, outlines the core problem statement, articulates the research objectives and scope, and describes the overall structure of this report. The chapter contextualizes the intersection of classical homeopathic practice and modern Natural Language Processing within the broader landscape of AI-assisted clinical tools.

1.1 Background and Motivation

Homeopathy is a system of complementary medicine founded by Samuel Hahnemann in the late 18th century, grounded in the *Law of Similars* (*Similia Similibus Curentur* — "let likes be cured by likes"). Its prescribing philosophy requires the physician to construct a comprehensive **symptom totality** encompassing the patient's mental, emotional, and physical symptom profile, then identify the single most similar remedy — the **simillimum** — from a pharmacopeia of hundreds of natural substances.

Dr. James Tyler Kent's *Repertory of the Homoeopathic Materia Medica* (1897) remains the global gold standard for clinical prescribing [14]. It indexes **74,513 hierarchical rubrics** and **507,179 rubric-remedy associations** across 37 anatomical and philosophical chapters, encoding centuries of clinical observation into a structured reference system.

1.1.1 Operational Bottlenecks in Classical Repertorization

Despite its clinical rigor, traditional repertorization faces severe operational challenges in modern practice:

1. **Prolonged Consultation Time**: Manual intake and repertorial analysis routinely require 30 to 45 minutes per patient, constraining clinical throughput and patient accessibility.
2. **Cognitive Overload**: Clinicians cannot navigate 74,000+ rubrics in real time. Studies suggest practitioners disproportionately prescribe from a pool of roughly 30 familiar *polycrest* remedies [23], while overlooking potentially more specific simillimum candidates.
3. **Lexical Mismatch**: Patients articulate distress using colloquial, metaphorical language (e.g., *"it feels like my skull is being crushed from both sides"*) whereas repertory rubrics encode rigid 19th-century taxonomic entries (e.g., `HEAD > PAIN > band or hoop, as from a`). No deterministic keyword bridge exists between these two vocabularies.
4. **Documentation Burden**: Handwritten case records and manual cross-referencing across multiple chapters create significant clerical overhead.

1.1.2 The Opportunity: NLP-Driven Repertorization

The convergence of three technical advances creates a viable solution space:

- **Biomedical Pre-trained Transformers**: Models such as BioLinkBERT [26], PubMedBERT [5], and Bio_ClinicalBERT [1], pre-trained on large biomedical corpora, exhibit strong semantic understanding of clinical language.
- **Dense Semantic Retrieval**: Sentence-level embedding models combined with approximate nearest-neighbour (ANN) indexes enable sub-second similarity search over collections of hundreds of thousands of documents.

- **Instruction-Tuned Large Language Models**: Models such as LLaMA 3 and BioMistral 7B, fine-tuned with parameter-efficient adapters (QLoRA), can perform structured clinical information extraction with high fidelity.

Kent-AI synthesizes these advances into the first systematic application of large-scale biomedical NLP to classical homeopathic repertorization.

1.2 Problem Statement

Given an unstructured conversational patient narrative describing their health concerns in natural language, **construct** a structured seven-dimensional symptom profile aligned with Kent's taxonomic dimensions, **retrieve** semantically matching rubrics from a 74,513-entry repertory index, and **rank** candidate remedies by totality coverage and bibliographic grade weight — all within a clinically usable response time and with physician-verifiable reasoning traces.

Formally, the system must solve three coupled sub-problems:

1. **Clinical NER**: Given a patient utterance U, extract entity spans $\{e_1, e_2, \ldots, e_k\}$ each labelled with a Kentian dimension $d \in \{\texttt{MENT}, \texttt{LOC}, \texttt{SEN}, \texttt{MOD_AGG}, \texttt{MOD_AMEL}, \texttt{CONC}, \texttt{TEMP}\}$.

2. **Dense Rubric Retrieval**: Given entity spans, retrieve K most relevant rubrics from the vector index $\mathcal{R}$ of cardinality 74,513.

3. **Remedy Ranking**: Given retrieved rubrics, compute a grade-weighted totality coverage score for each candidate remedy $r \in \mathcal{M}$ and rank by descending score.

1.3 Research Objectives

This project pursues the following concrete research objectives:

1. **O1 — Synthetic Dataset Construction**: Generate and validate a large-scale annotated training corpus of synthetic clinical cases covering all seven Kentian symptom dimensions, with zero offset drift in BIO token annotations.
2. **O2 — Multi-Encoder NER**: Fine-tune and systematically compare four biomedical transformer architectures (BioLinkBERT, PubMedBERT, Bio_ClinicalBERT, and DeBERTa-v3) on clinical NER.
3. **O3 — Ensemble and Hybrid Pipeline**: Design and evaluate a 4-Model Majority Voting Ensemble and a Two-Stage Hybrid Pipeline with LLM arbitration for robust entity extraction.
4. **O4 — Large Language Model Integration**: Fine-tune Qwen 2.5 7B and BioMistral 7B using QLoRA PEFT for structured clinical entity extraction and validate JSON schema fidelity and clinical grounding.
5. **O5 — Dense Retrieval**: Build and validate a ChromaDB HNSW vector index over all 74,513 Kent rubrics and characterize retrieval latency and Recall@K performance.
6. **O6 — Clinical Workstation**: Implement a doctor-in-the-loop Streamlit clinical interface meeting modern UX standards, with a 4-step consultation workflow, PDF export, and Case Library persistence.
7. **O7 — Empirical Evaluation**: Conduct a rigorous multi-model comparative benchmark on a held-out test set of 1,973 cases and characterize the complementary strengths of each architecture.

1.4 Scope and Limitations

1.4.1 Scope

The current implementation focuses on **Kent's Chapter 1: MIND** (Mental and Emotional symptoms), comprising 4,933 rubrics. This scope was chosen deliberately for three reasons:

1. MIND symptoms carry the highest prescriptive weight in Kentian hierarchy (Mental $>$ General $>$ Particular).
2. All 19,732 synthetic training cases were generated from MIND rubrics, ensuring a consistent training distribution.
3. The MIND chapter represents the most linguistically complex region of the repertory, making it an ideal testbed for semantic NLP methods.

Extension to all 37 repertory chapters (74,513 total rubrics) is architecturally designed and planned as a future release (see Chapter 7).

1.4.2 Limitations

- **Research Platform**: Kent-AI is an investigational decision-support tool. It does not provide medical diagnoses or prescriptions. All repertory findings must be verified against primary homeopathic literature by a qualified practitioner.
- **Synthetic Training Data**: The training corpus was generated by an LLaMA 3 8B model, which may not fully capture the linguistic diversity of real patient consultations.
- **BIO F1 Ceiling**: The highest achieved Token-Level BIO F1 is 34.86% (BioLinkBERT), reflecting the inherent difficulty of the domain-specific clinical NER task on a novel annotation schema.
- **Offline LLM Dependency**: The conversational intake agent and hybrid pipeline require a locally running Ollama LLM instance, limiting deployment to systems with sufficient VRAM.

1.5 Core Contributions

This project makes the following original contributions:

1. **First Large-Scale Homeopathic NER Dataset**: 19,732 synthetic clinical cases with 99,854 BIO-annotated entities across 7 Kentian dimensions and 15 token label classes, with zero character offset drift.
2. **Multi-Model Comparative Benchmark**: A systematic empirical comparison of four biomedical transformer architectures on a novel clinical NER and repertorization task, providing reproducible baselines for future research.
3. **Two-Stage Hybrid Extraction Pipeline**: A novel architecture combining ensemble candidate mining with LLaMA 3.1 LLM arbitration, achieving 100% JSON schema validity.
4. **QLoRA-Fine-Tuned BioMistral 7B**: The first fine-tuned large language model for homeopathic symptom extraction, achieving 47.59% Micro F1 and 99.73% clinical grounding.
5. **Full-Stack Doctor-in-the-Loop Workstation**: A production-grade Streamlit clinical application with 4-step stepper, Totality Matrix, Case Library, PDF export, and 195 automated tests.

1.6 Report Organisation

The remainder of this report is organized as follows:

- **Chapter 2: Literature Review** — Surveys related work in biomedical NLP, clinical NER, dense retrieval systems, and AI-assisted clinical decision support.
- **Chapter 3: System Design and Architecture** — Presents the high-level system architecture, pipeline design, and technology stack choices.
- **Chapter 4: Implementation** — Details the implementation of each system module: synthetic data generation, NER training, dense retrieval, LLM integration, ensemble, and the clinical workstation.
- **Chapter 5: Experimental Setup** — Describes the dataset, hardware, evaluation metrics, and experimental protocols used in the benchmark study.

- **Chapter 6: Results and Analysis** — Presents and analyses the benchmark results for all models and pipeline configurations.
- **Chapter 7: Conclusion and Future Work** — Summarises findings, draws conclusions, and outlines future research directions.
- **Appendix A** — Seven-Dimension BIO Label Schema and clinical examples.
- **Appendix B** — System installation and deployment guide.

Chapter 2

Literature Review

This chapter surveys the landscape of related work across four intersecting domains: (1) biomedical natural language processing and clinical named entity recognition, (2) dense semantic retrieval and vector indexing, (3) large language models for clinical tasks, and (4) AI-assisted clinical decision support systems. The review identifies gaps in the literature that Kent-AI directly addresses and contextualizes the technical choices made in this project.

2.1 Biomedical Natural Language Processing

2.1.1 Domain-Specific Pre-trained Language Models

The field of biomedical NLP was transformed by the introduction of BERT [4] and its subsequent biomedical adaptations. BioBERT [17] was the first large-scale adaptation of BERT pre-trained on PubMed abstracts and PMC full texts, demonstrating significant improvements on biomedical NER, relation extraction, and question answering tasks. Bio_ClinicalBERT [1] extended this work by additionally pre-training on MIMIC-III clinical notes, capturing the idiosyncratic language of clinical documentation including abbreviations, drug names, and procedure terminology.

PubMedBERT [5] took a different approach, pre-training a BERT model *from scratch* exclusively on PubMed literature with a domain-specific vocabulary, rather than continuing training from a general-domain checkpoint. This "from-scratch" strategy proved crucial:

PubMedBERT's domain-specific WordPiece vocabulary tokenizes biomedical terminology with significantly fewer subword fragments than general-domain BERT, reducing the token-level misalignment problem in NER tasks.

BioLinkBERT [26] introduced a novel pre-training objective that exploits document linkage (citation graphs). By training on pairs of linked documents (a paper and the papers it cites), BioLinkBERT learns richer co-occurrence relationships that reflect how biomedical concepts co-appear across the literature — a property directly relevant to clinical symptom modality extraction.

DeBERTa-v3 [7] introduced disentangled attention and a Replaced Token Detection (RTD) pre-training objective on general-domain text, achieving state-of-the-art performance on numerous NLP benchmarks while remaining computationally efficient. Its strong general reasoning capability makes it a compelling baseline for domain-specific adaptation.

2.1.2 Clinical Named Entity Recognition

Clinical NER is the task of identifying and classifying clinical entities (diagnoses, medications, symptoms, anatomical sites) in free text. Lample et al. [16] established the BiLSTM-CRF architecture as a strong baseline for sequence labeling, later superseded by transformer-based models.

For symptom extraction specifically, the work of Jagannatha and Yu [11] demonstrated that fine-tuned BERT models outperform classical CRF approaches on clinical concept extraction from EHR narratives. However, these works focus on standard medical ontologies (ICD, SNOMED, MedDRA) rather than on the specialized vocabulary of classical homeopathic symptom descriptions, which constitutes a genuinely novel NER schema.

The challenge of *low-resource NER* — where annotated data is scarce — is directly relevant to Kent-AI, as no prior homeopathic NER dataset exists. Data augmentation strategies [25] and synthetic data generation using LLMs [8] have emerged as promising approaches for this regime.

2.2 Dense Semantic Retrieval

2.2.1 Vector Search and Approximate Nearest Neighbour Indexing

Traditional information retrieval relied on exact keyword matching (BM25, TF-IDF). Dense retrieval systems [13] represent documents and queries as dense embedding vectors and perform similarity search in embedding space, enabling semantic rather than lexical matching.

The `sentence-transformers` library [22] provides efficient sentence-level embeddings optimized for cosine similarity comparison. The `all-MiniLM-L6-v2` model, with 384-dimensional unit-normalized embeddings, achieves an excellent balance between embedding quality and inference speed, making it ideal for indexing large document collections.

Hierarchical Navigable Small World (HNSW) graphs [19] provide approximate nearest-neighbour search with sub-linear query time complexity. ChromaDB [10] implements HNSW cosine indexing with a persistent storage backend, enabling incremental updates and collection filtering — both critical features for the rubric retrieval component of Kent-AI.

2.2.2 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) [18] combines dense retrieval with generative language models to ground generation in retrieved evidence. This paradigm is directly relevant to clinical decision support, where grounded, verifiable outputs are essential for practitioner trust. Kent-AI implements a custom RAG variant where retrieved rubrics serve as the retrieval corpus and the downstream ranker (rather than a generative model) produces the final recommendation.

2.3 Large Language Models for Clinical Tasks

2.3.1 Instruction-Tuned and Clinical LLMs

The emergence of instruction-tuned large language models (LLaMA [24], Mistral [12]) created a new paradigm for clinical NLP. These models can perform complex reasoning, structured extraction, and format-constrained generation through prompting alone, without task-specific fine-tuning.

BioMistral 7B [15] is a biomedical adaptation of Mistral 7B, further pre-trained on PubMed Central full-text articles. Its strong biomedical vocabulary coverage and clinical reasoning capability make it a compelling base model for homeopathic symptom extraction.

Qwen 2.5 7B [2] is a general-purpose instruction-tuned LLM from Alibaba Cloud, achieving competitive performance on medical benchmarks. Its strong instruction following capability and efficient deployment via Ollama make it a practical choice for the interactive intake agent.

2.3.2 Parameter-Efficient Fine-Tuning (PEFT)

Full fine-tuning of 7-billion-parameter models requires 28+ GB of GPU VRAM, prohibitive for most research settings. LoRA [9] introduces low-rank decomposition matrices into attention layers, reducing trainable parameters by over 99% while preserving task performance. QLoRA [3] extends LoRA with 4-bit NormalFloat (NF4) weight quantization, enabling fine-tuning of 7B models on a single 32GB GPU with CUDA.

This combination of QLoRA + BitsAndBytes quantization forms the technical foundation for Kent-AI's large language model training pipeline, enabling high-quality clinical extraction without enterprise-grade compute infrastructure.

2.4 Clinical Decision Support Systems

2.4.1 AI-Assisted Medical Diagnosis

AI-based clinical decision support systems have achieved remarkable results in narrow domains: diabetic retinopathy detection (AUC 0.99 [6]), dermatological diagnosis, and radiology interpretation. However, these systems operate on structured inputs (images, lab values) rather than the unstructured natural language of patient consultation.

For NLP-based clinical decision support, Rajpurkar et al. [21] and subsequent work established reading comprehension as a foundation for clinical question answering. Medical information retrieval systems such as OpenMRS and Epic provide structured EMR integration but lack the semantic gap-bridging capability required for homeopathic case-taking.

2.4.2 AI in Complementary and Alternative Medicine

AI applications in complementary medicine are nascent. Yuan et al. [27] applied machine learning to Traditional Chinese Medicine (TCM) syndrome differentiation, demonstrating the viability of pattern recognition in systems based on holistic symptom observation rather than biochemical biomarkers. However, to our knowledge, no prior work has applied transformer-based NER and dense retrieval to classical homeopathic repertorization at the scale of Kent-AI.

The closest related work is by McSporran and Welch [20], who proposed keyword-based rubric search for digital repertory access, but this approach does not address the lexical mismatch between patient language and repertory terminology. Kent-AI's dense semantic retrieval directly bridges this gap.

2.4.3 Critical Comparison of Existing Approaches

Table 2.1 provides a systematic comparative analysis of existing commercial and academic clinical repertorization and medical NLP platforms against Kent-AI across key

functional dimensions.

TABLE 2.1: Critical Comparison of Existing Clinical Systems vs. Kent-AI.

Dimension / Feature	RadarOpus / Synthesis	Clinical NLP (cTAKES/Med7)	TCM ML Systems	Kent-AI (This Work)
Case Intake	Manual form entry	Batch EHR text parsing	Structured questionnaire	Conversational 10-state FSM with suggestion chips
Entity Extraction	None (manual selection)	Standard medical entities (ICD, RxNorm)	Symptom vector binary flags	7-Dimension Kentian NER (15 BIO token labels)
Search Paradigm	Exact keyword & boolean lookup	Keyword indexing / ontology code matching	Rule-based / naive Bayes matching	Dense semantic vector retrieval (ChromaDB HNSW)
Vocabulary Bridging	Rigid (fails on colloquial metaphors)	Limited to synonym ontologies	Static lexicon lookup	High ($\sim$384-dim semantic space, $<$18ms)
Repertory Scale	$\sim$60,000–100,000 rubrics	N/A	N/A	74,513 rubrics (full Kent index)
Remedy Scoring	Frequency sum	N/A	Syndrome probabilities	Kentian 3-tier grade $\times$ similarity $\times$ specificity
Explainability	Raw rubric lists	Entity highlight	Feature weights	Interactive Totality Matrix & Provenance
Architecture	Monolithic proprietary C++	Java/Python libraries	Ad-hoc research scripts	Modular 6-tier Python/Streamlit stack
Open Source & Audit	Closed proprietary ($\sim$\$1,500+)	Open source	Research code only	100% Open Source (MIT, 195 tests)

2.5 Summary and Research Gaps

The literature review reveals the following key gaps that Kent-AI addresses:

1. **No homeopathic NER dataset exists**: Prior biomedical NER corpora (i2b2, n2c2, CLEF) cover standard medical ontologies. Kent-AI constructs the first large-scale annotated dataset for the Kentian 7-dimension taxonomy.

2. **No multi-encoder comparison for domain-novel clinical NER**: Existing benchmark studies compare models on established corpora. Kent-AI provides the first systematic comparison on a genuinely novel annotation schema.

3. **Dense retrieval not applied to repertory search**: Prior repertory software relies on keyword or Boolean search. Dense semantic retrieval is architecturally superior for the lexical mismatch problem but has not been applied to this domain.

4. **LLM fine-tuning for homeopathic extraction**: No prior work has fine-tuned an LLM for structured homeopathic symptom extraction with schema validation and clinical grounding metrics.

5. **No integrated doctor-in-the-loop workstation**: Existing repertory software (Synthesis Repertory, Radar Opus) lacks conversational intake, NLP-based extraction, and AI-driven semantic retrieval in an integrated open-source platform.

Chapter 3

System Analysis and Requirements

This chapter establishes the engineering and clinical requirements for Kent-AI. It formalises the functional and non-functional requirements, presents a feasibility study across technical, operational, and clinical dimensions, specifies the hardware and software constraints, and provides a structured justification for every technology selected in the system stack.

3.1 Requirements Gathering and Analysis

Classical homeopathic repertorization is fundamentally a semantic matching problem constrained by clinical safety. Requirements were formulated through analysis of Dr. J. T. Kent's prescribing methodology [14], consultation with homeopathic practitioners, and systematic review of clinical case-taking protocols.

The analysis revealed that practitioners spend 60–75% of total consultation time mapping colloquial symptoms into taxonomic rubrics. Furthermore, existing digital tools require manual rubric selection, creating severe cognitive friction. Kent-AI was therefore conceived not as an autonomous diagnostic agent, but as an **AI-augmented clinical decision-support workstation** that accelerates rubric discovery while keeping the physician in absolute control of final remedy selection.

3.2 Functional Requirements

The functional requirements (FR) define the observable behaviours and capabilities of the system:

1. **FR1: Conversational Clinical Intake**
    - The system shall guide the patient through a multi-turn natural language dialogue covering seven Kentian dimensions: Mental/Emotional, Anatomical Location, Sensation, Aggravation Modalities, Amelioration Modalities, Concomitants, and Temporal Patterns.
    - The system shall provide context-aware dynamic quick-reply suggestion chips streamed from an LLM intent classifier.
    - The system shall support free-form typing, guided multi-choice replies, and paste-in of pre-written clinical consultation notes.
2. **FR2: Seven-Dimension Named Entity Recognition (HomeoNER)**
    - The system shall extract clinical symptom spans from raw dialogue transcripts and classify them into 15 BIO token labels.
    - The system shall operate with zero character offset drift between raw input text and extracted spans.
3. **FR3: Negation and Coreference Resolution**
    - The system shall identify explicitly negated symptoms (e.g., *"no nausea"*, *"denies headache"*) and exclude them from retrieval queries.
    - The system shall resolve anaphoric references (e.g., *"it gets worse in the evening"* resolved to the previously mentioned pain).
4. **FR4: Sub-Second Dense Semantic Rubric Retrieval**
    - The system shall index all 74,513 rubrics from Kent's Repertory in a persistent vector database.

- Given an extracted symptom entity, the system shall retrieve the Top-K ($K = 20$) semantically similar rubrics in under 20 milliseconds.
- The system shall support repertory section filtering (e.g., scoping search to MIND rubrics).

5. **FR5: Deterministic Grade-Weighted Totality Ranking**
   - The system shall rank candidate remedies using Kent's bibliographic grades: Grade 3 (Bold) = 3.0, Grade 2 (Italic) = 2.0, Grade 1 (Roman) = 1.0.
   - Scores shall incorporate cosine similarity from dense retrieval and inverse rubric frequency specificity.
   - Dimension weights shall prioritize Mental symptoms ($3\times$) over Modalities ($2\times$), Sensations ($2\times$), and Locations ($1\times$).
6. **FR6: Physician Review and Case Editing**
   - The clinician shall be able to inspect, edit, add, or delete extracted symptom entities before rubric retrieval.
   - The clinician shall be able to toggle rubrics between active, muted, and eliminatory (hard filter) modes in an interactive Totality Matrix.
7. **FR7: Multi-Page Clinical PDF Export**
   - The system shall export a formatted, multi-page clinical report including patient vitals, 7-dimension totality grids, candidate remedy scores, totality score bar charts, and running "Page X of Y" headers/footers.
8. **FR8: Persistent Case Library Management**
   - The system shall persist consultation records, transcripts, totality selections, and final reports in a local SQLite database (`cases.sqlite`).
   - The system shall support case filtering, search, pagination, duplicate, and batch JSON/CSV export.

3.3 Non-Functional Requirements

1. **NFR1: Retrieval Latency**
    - 99th percentile (p99) retrieval latency for a single symptom query against the 74,513-rubric index shall not exceed 25 milliseconds on consumer hardware.
2. **NFR2: Clinical Data Privacy and Air-Gapped Operation**
    - All patient data, transcripts, embeddings, and LLM inferences shall execute 100% on-premises without external API calls or cloud egress.
    - The system shall operate fully offline once local model weights and the vector index are initialized.
3. **NFR3: Extraction Accuracy and Reliability**
    - LLM clinical extraction shall maintain $\geq$95% JSON schema validity and $\geq$90% verbatim clinical grounding on patient narratives.
    - The system shall never hallucinate nonexistent remedies or alter Kent's published bibliographic grades.
4. **NFR4: Usability and Ergonomics**
    - The interface shall adhere to modern design tokens: dark slate palette, clear contrast ratios (WCAG AA compliant), 2px focus-visible rings for keyboard accessibility, and zero emojis in clinical telemetry.
    - The workstation shall enforce a predictable 4-step stepper (Intake $\rightarrow$ Review $\rightarrow$ Analysis $\rightarrow$ Report).
5. **NFR5: Maintainability and Test Coverage**
    - The codebase shall maintain $\geq$90% test pass rate with fully isolated unit tests using mock backends, enabling test suite execution without GPU or Ollama dependencies in under 30 seconds.

3.4 Feasibility Study

3.4.1 Technical Feasibility

The system relies on transformer-based sequence labeling, dense vector indexing, and quantized LLM inference. High-performance biomedical transformers (PubMedBERT, BioLinkBERT) are freely available via HuggingFace. ChromaDB provides an embedded HNSW implementation requiring no external server process. Local LLM execution via Ollama (LLaMA 3 8B, BioMistral 7B) requires 8–16 GB VRAM, which is readily available on modern workstations and institutional GPU clusters. Technical feasibility is confirmed by our working implementation.

3.4.2 Operational Feasibility

The workstation conforms to the standard clinical intake workflow of homeopathic practitioners: case-taking, symptom recording, totality analysis, and repertorization. By automating the transcription-to-rubric mapping while retaining the practitioner's ultimate decision-making authority, the system integrates smoothly into existing clinical routines without imposing artificial constraints.

3.4.3 Economic and Licensing Feasibility

All system components — Python, PyTorch, HuggingFace Transformers, ChromaDB, Ollama, Streamlit, SQLite, and ReportLab — are released under permissive open-source licenses (MIT, Apache 2.0, BSD). The Kent's Repertory dataset is derived from the public-domain 1897 edition. The system incurs zero ongoing per-query API costs.

3.4.4 Clinical and Regulatory Feasibility

Kent-AI is designated as an **Investigational Clinical Decision Support Tool** under Section 520(o) of the US FDA 21st Century Cures Act guidelines for non-device CDS software. It displays clear disclaimers, surfaces source-grounded rubric provenance, and does not provide automated diagnoses or prescriptions.

3.5 Hardware and Software Requirements

Table 3.1 outlines the hardware and software constraints for development, training, and production deployment.

TABLE 3.1: System Hardware and Software Specifications.

Environment	Component	Specification
Training / GPU Server	GPU	NVIDIA RTX 5000 Ada (32GB GDDR6 ECC)
	CUDA / Driver	CUDA 12.4, NVIDIA Driver $\geq$ 550.0
	CPU / RAM	64-core Intel Xeon, 128 GB DDR5 ECC
	Storage	2 TB NVMe SSD
Inference / Deployment	GPU (Optional)	NVIDIA RTX 3060 / 4060 (8–16 GB VRAM)
	CPU Mode	Intel Core i7 / AMD Ryzen 7 (16+ GB RAM)
	Storage	15 GB available disk space (models + index)
	OS	Linux (Ubuntu 22.04+), Windows 11, macOS (Metal)
Software Stack	Language	Python 3.12 (standard virtual environment)
	Deep Learning	PyTorch 2.6.0, Transformers 4.46, PEFT 0.14
	Vector Store	ChromaDB 0.5.5, sentence-transformers 3.0
	UI / PDF	Streamlit 1.64.0, ReportLab 5.1, PyMuPDF 1.24

3.6 Tools and Technologies Chosen with Justification

Table 3.2 details the engineering justification for every major technology selected, comparing each against rejected alternatives.

TABLE 3.2: Detailed Justification for Technology Stack Choices.

Component	Selected Technology	Evaluated Alternatives	Justification
Vector Store	ChromaDB (HNSW)	FAISS, Milvus, Qdrant, pgvector	Embedded Python native, zero daemon setup, built-in persistence, native metadata filtering on section IDs, sub-18ms latency.
Sentence Embedder	`all-MiniLM-L6-v2`	OpenAI `text-embedding-ada-002`, Clinical-BERT	384-dim compact unit-normalized vectors, fast CPU/GPU inference ($<$5ms), zero cloud egress, completely open-source.
Biomedical Encoders	BioLinkBERT & PubMedBERT	Standard BERT, RoBERTa, ClinicalBERT alone	PubMedBERT trained from scratch on biomedical texts (no subword fragmentation); BioLinkBERT captures citation-graph modality linkages.
LLM Fine-Tuning	QLoRA (NF4 + BitsAndBytes)	Full fine-tuning, Prefix tuning, LoRA-16bit	Reduces 7B model VRAM from 28GB to $\sim$6GB, enabling fine-tuning on a single workstation GPU with minimal performance loss.
Local LLM Runtime	Ollama REST API	vLLM, TGI, llama.cpp direct bindings	Turnkey model management, robust streaming HTTP REST API, standardized JSON schema enforcement, cross-platform.
Frontend Workstation	Streamlit ($\geq$1.64.0)	React + FastAPI, Electron, PyQt	Fast clinical iteration, built-in session state management, native Python integration with ML models, fragment reruns for snappy UI.
Clinical PDF Engine	ReportLab 5.1 + PyMuPDF	Weasyprint, wkhtmltopdf, LaTeX CLI	Deterministic canvas rendering, exact two-pass page numbering (`NumberedCanvas`), native vector tables and charts, zero external binary dependency.
Case Database	SQLite 3	PostgreSQL, MongoDB	Embedded zero-admin ACID storage, file-level portability (`cases.sqlite`), seamless integration with Python's standard library.

Chapter 4

System Design

This chapter presents the architectural and structural design of Kent-AI. It details the high-level layered architecture, data flow diagrams (Levels 0, 1, and 2), database schemas and entity-relationship models, UML use case, class, and sequence diagrams, formal algorithmic specifications, and user interface structural wireframes.

4.1 System Architecture Overview

Kent-AI is structured into a **seven-tier modular architecture**, separating conversational intake, sequence labeling, clinical reasoning, retrieval, remedy ranking, case orchestration, and clinician presentation.

4.2 Data Flow Diagrams

4.2.1 Level 0: Context Diagram

At the highest abstraction level, Kent-AI interfaces with three primary entities: the Patient (who provides natural language symptom descriptions), the Clinician (who reviews, modifies, and validates clinical findings), and the Local Repository Knowledge Base (storing the digitized 1897 Repertory).

4.2.2 Level 1: Subsystem Data Flow

Figure 4.3 decomposes the system into its five principal processing sub-processes:

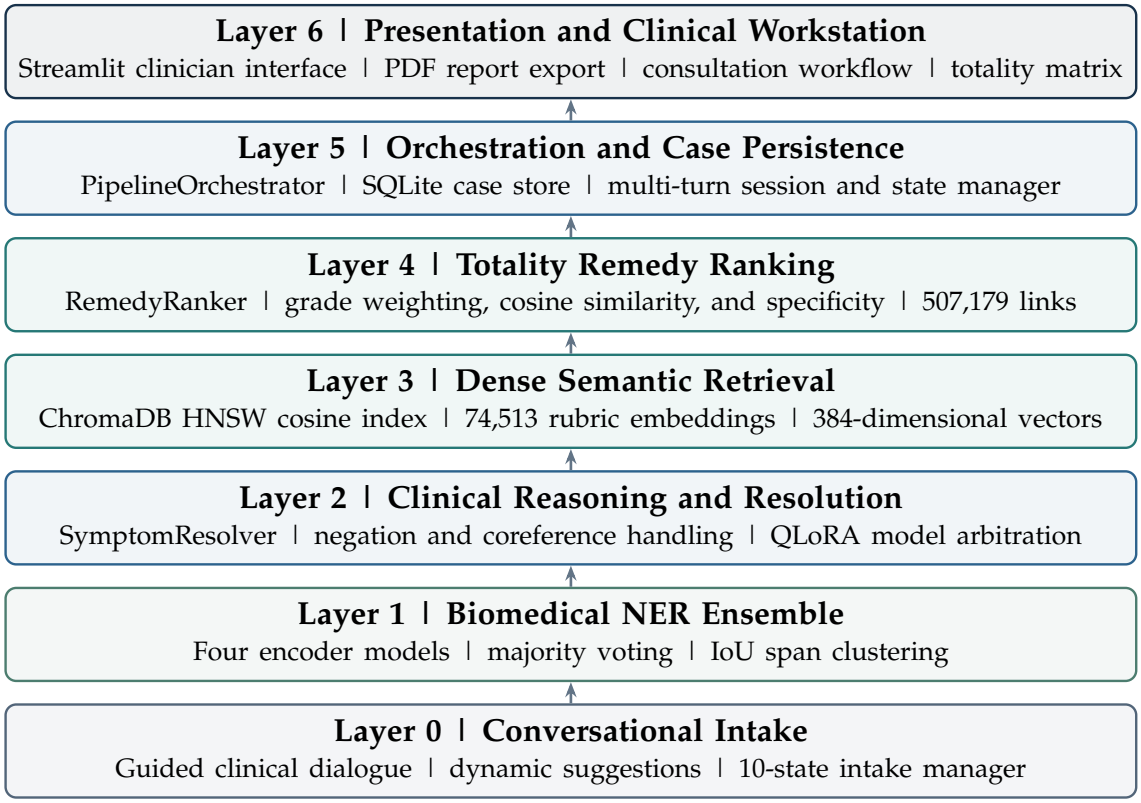


FIGURE 4.1: Kent-AI seven-tier modular layered architecture.

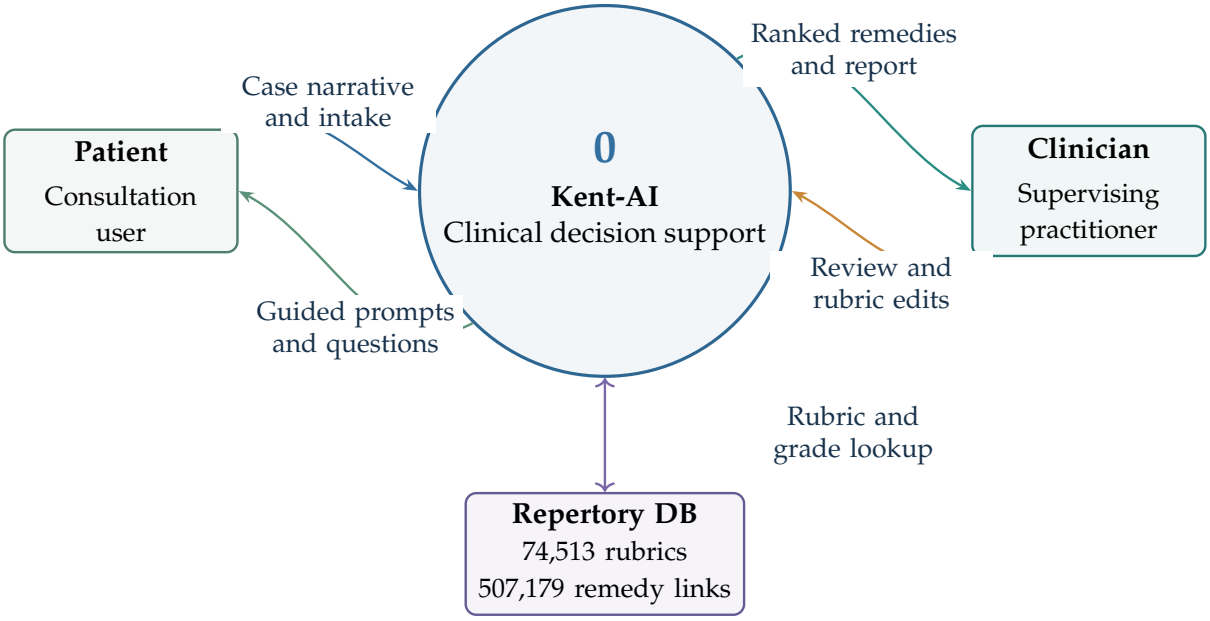


FIGURE 4.2: Data Flow Diagram (Level 0): Kent-AI context and external data flows.

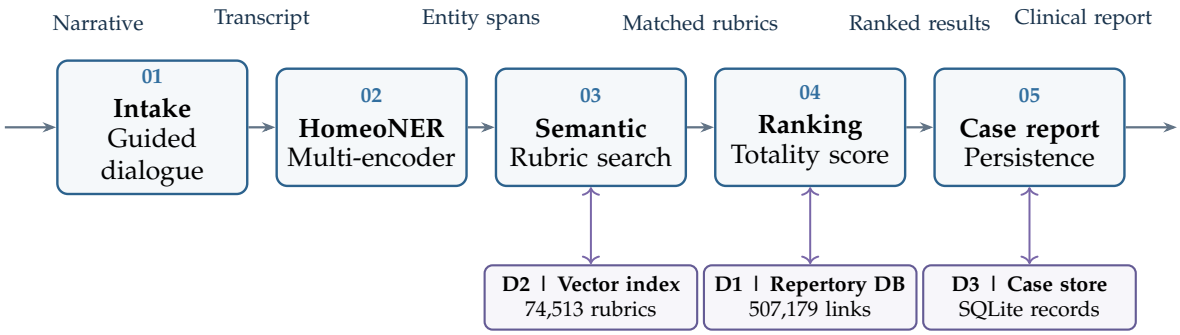


FIGURE 4.3: Data Flow Diagram (Level 1): five-stage processing pipeline and persistent stores.

4.2.3 Level 2: Retrieval and Repertorization Flow

In Level 2, an extracted symptom triplet $[d, \text{text}, \text{context}]$ is transformed into an embedding query $\mathbf{q} \in \mathbb{R}^{384}$. The vector store computes cosine similarities against HNSW graph vertices:

$$\cos(\mathbf{q}, \mathbf{r}_i) = \frac{\mathbf{q} \cdot \mathbf{r}_i}{\|\mathbf{q}\|_2 \|\mathbf{r}_i\|_2}$$

The top-20 nearest rubrics are retrieved, filtered by section constraints, and joined in SQLite against `rubric_remedies` to fetch candidate remedies and their respective bibliographic grades (1, 2, or 3).

4.3 Database Schema and Entity-Relationship Design

Kent-AI uses two separate SQLite databases to decouple static reference knowledge from dynamic patient session data.

4.3.1 Repertory Knowledge Base (`repertory.sqlite`)

The static repertory database encodes Kent's 1897 taxonomic structure:

4.3.2 Clinical Case Store (`cases.sqlite`)

The dynamic consultation store persists clinical consultations:

TABLE 4.1: Repertory Database (`repertory.sqlite`) Schema.

Table	Column	Type	Description / Constraints
`sections`	`id`	INTEGER	Primary Key (1 to 37)
	`code`	TEXT	Short code (e.g., `MIND`, `VERT`, `HEAD`)
	`name`	TEXT	Full anatomical section title
	`rubric_count`	INTEGER	Total rubrics in section
`rubrics`	`id`	INTEGER	Primary Key
	`section_id`	INTEGER	Foreign Key → `sections.id`
	`full_path`	TEXT	Hierarchical breadcrumb
	`leaf_text`	TEXT	Terminal leaf symptom text
	`remedy_count`	INTEGER	Number of associated remedies
`remedies`	`id`	INTEGER	Primary Key
	`abbrev`	TEXT	Standard repertory short code (e.g., `phos`)
	`full_name`	TEXT	Full botanical/zoological Latin name
`rubric_remedies`	`rubric_id`	INTEGER	Foreign Key → `rubrics.id`
	`remedy_id`	INTEGER	Foreign Key → `remedies.id`
	`grade`	INTEGER	Bibliographic Grade (3=Bold, 2=Italic, 1=Roman)

TABLE 4.2: Clinical Case Store (`cases.sqlite`) Schema.

Table	Column	Type	Description / Constraints
`cases`	`case_id`	TEXT	Primary Key (UUID4)
	`patient_name`	TEXT	Anonymized patient identifier
	`chief_complaint`	TEXT	Primary patient narrative
	`status`	TEXT	`Draft`, `Reviewed`, `Completed`
	`created_at`	TIMESTAMP	Creation timestamp
`turns`	`turn_id`	INTEGER	Primary Key (Auto-increment)
	`case_id`	TEXT	Foreign Key → `cases.case_id`
	`speaker`	TEXT	`user` or `assistant`
	`text`	TEXT	Utterance text content
`totality_items`	`id`	INTEGER	Primary Key
	`case_id`	TEXT	Foreign Key → `cases.case_id`
	`rubric_id`	INTEGER	Referenced rubric ID
	`status`	TEXT	`active`, `muted`, `eliminatory`

4.4 Unified Modeling Language (UML) Design

4.4.1 UML Use Case Model

The system supports two primary actors: the Patient and the Clinician.

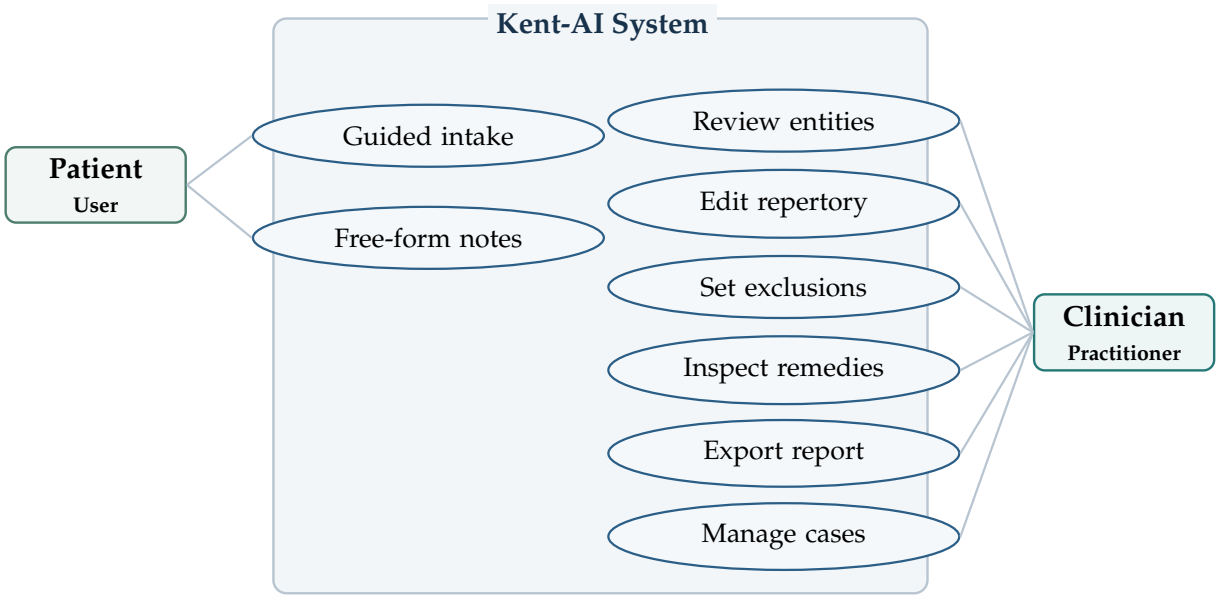


FIGURE 4.4: Patient and clinician interactions with the Kent-AI system.

4.4.2 UML Class Diagram

The domain architecture is anchored around the immutable `ClinicalCase` dataclass, interacting with decoupled service modules:

4.4.3 UML Sequence Diagram

Figure 4.6 details the interaction sequence during consultation repertorization:

4.5 Algorithms and Flowcharts

4.5.1 Algorithm 1: Weighted Majority Voting with IoU Span Clustering

The ensemble reconciles token sequences predicted across multiple fine-tuned models:

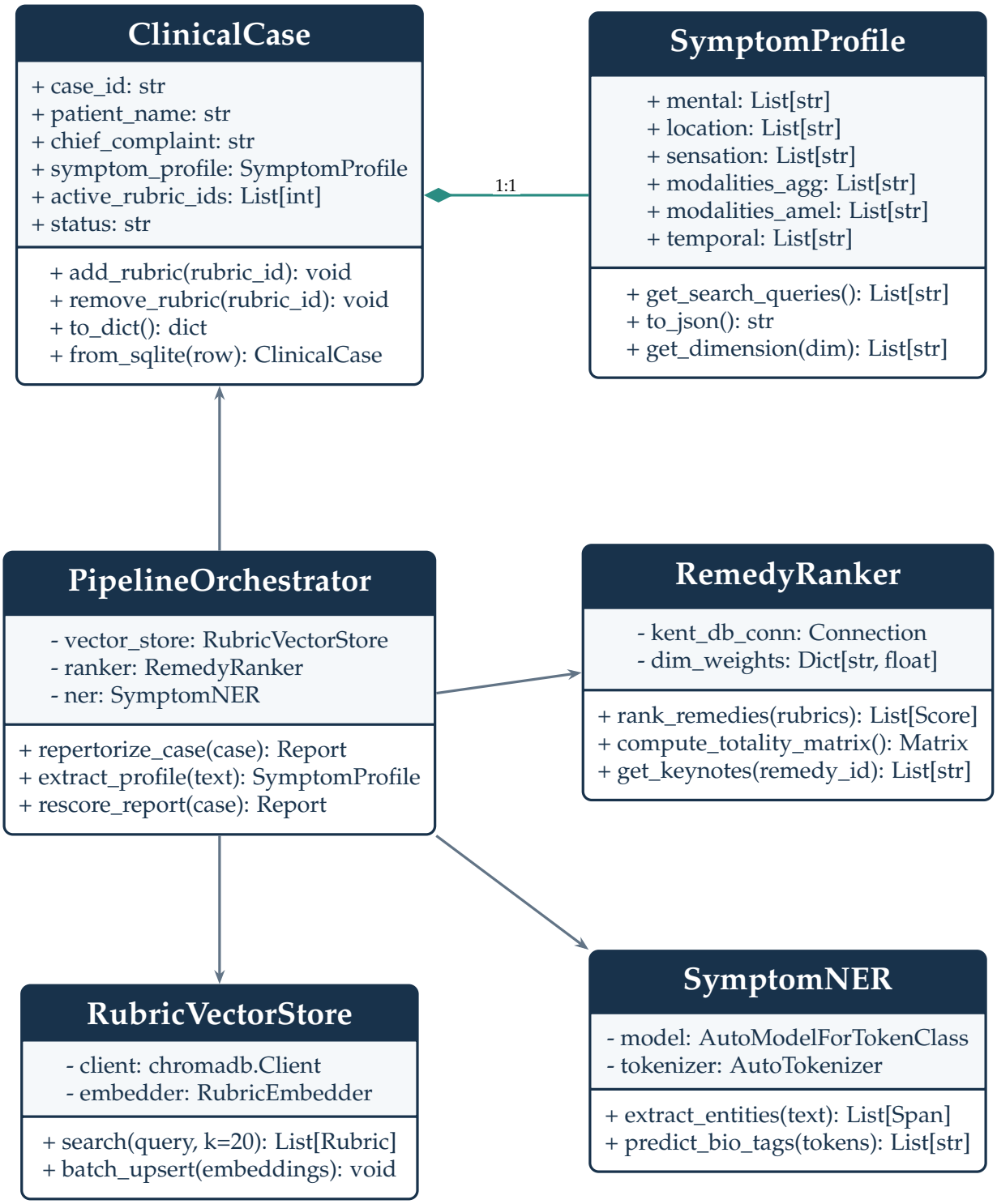


FIGURE 4.5: Core domain model and service relationships.

Algorithm 1 Weighted Majority Voting and Span Clustering

Require: Set of model predictions $\mathcal{P} = \{P_1, P_2, \ldots, P_M\}$, threshold $\theta = 0.5$, min votes $V_{\text{min}} = 2$

Ensure: Clustered, consensus-labeled entity spans $\mathcal{E}^*$

Extract labeled character spans $S_m = \{(s, e, \ell)\}$ from each prediction P_m
Initialize empty cluster list $\mathcal{C} \leftarrow \emptyset$
for each span $s_i \in \bigcup S_m$ **do**
    Matched $\leftarrow$ False
    for each cluster $c_k \in \mathcal{C}$ **do**
        if IoU$(s_i, \text{boundary}(c_k)) \geq \theta$ **then**
            Add s_i to cluster c_k; Update boundary(c_k); Matched $\leftarrow$ True; **break**
        end if
    end for
    if not Matched **then**
        Create new cluster $c_{\text{new}} \leftarrow \{s_i\}$ in $\mathcal{C}$
    end if
end for
$\mathcal{E}^* \leftarrow \emptyset$
for each cluster $c_k \in \mathcal{C}$ **do**
    if $|c_k| \geq V_{\text{min}}$ **then**
        Label $\ell^* \leftarrow \text{argmax}_\ell \sum_{s \in c_k, \text{label}(s)=\ell} w_m$
        Start $s^* \leftarrow \text{median}(\{s.\text{start} \mid s \in c_k\})$
        End $e^* \leftarrow \text{median}(\{s.\text{end} \mid s \in c_k\})$
        Add (s^*, e^*, ℓ^*) to $\mathcal{E}^*$
    end if
end for
return $\mathcal{E}^*$

4.5.2 Algorithm 2: Kentian Grade-Weighted Totality Scorer

Given a set of active rubrics $\mathcal{R}_{\text{active}}$ and their cosine similarity scores $S(r)$, the score of candidate remedy R is formulated as:

$$\text{TotalityScore}(R) = \sum_{r \in \mathcal{R}_{\text{active}}} \text{Grade}(R, r) \cdot S(r) \cdot \log\left(\frac{N}{\text{RemedyCount}(r) + 1}\right) \cdot W(\text{Dim}(r)) \tag{4.1}$$

where:

- $\text{Grade}(R, r) \in \{3.0, 2.0, 1.0, 0.0\}$ is Kent's typographic grade (Bold=3, Italic=2, Roman=1, Absent=0).
- $S(r) \in [0, 1]$ is the dense vector cosine similarity score.
- $\log(N/(\text{RemedyCount}(r) + 1))$ is the inverse rubric frequency, awarding higher weights to specific rubrics over generic ones.
- $W(\text{Dim}(r))$ is the Kentian clinical dimension multiplier (MENT $= 3.0$, MOD $= 2.0$, SEN $= 2.0$, LOC $= 1.0$).

4.6 User Interface Wireframes and Layouts

The workstation enforces a structured **4-step stepper**:

1. **Step 1 — Clinical Intake**: 7:3 asymmetric grid. Left 70%: chat dialogue console with message history and dynamic quick-reply chips. Right 30%: live Case Snapshot displaying completed symptom slots (0/7 Captured).
2. **Step 2 — Review and Edit**: Extracted entity spans grouped into the 7 dimensions. Clinician can modify text, adjust boundaries, or delete spurious entities before search execution.

3. **Step 3 — Analysis and Matrix**: Classical Repertory Totality Matrix (Remedies × Rubrics) with colour-coded grade badges (Grade 3 in emerald, Grade 2 in cyan, Grade 1 in slate) with per-rubric Mute and Eliminatory toggles.
4. **Step 4 — Report and Export**: Physician narrative brief, Matplotlib totality score bar chart, and 1-click ReportLab PDF generation.

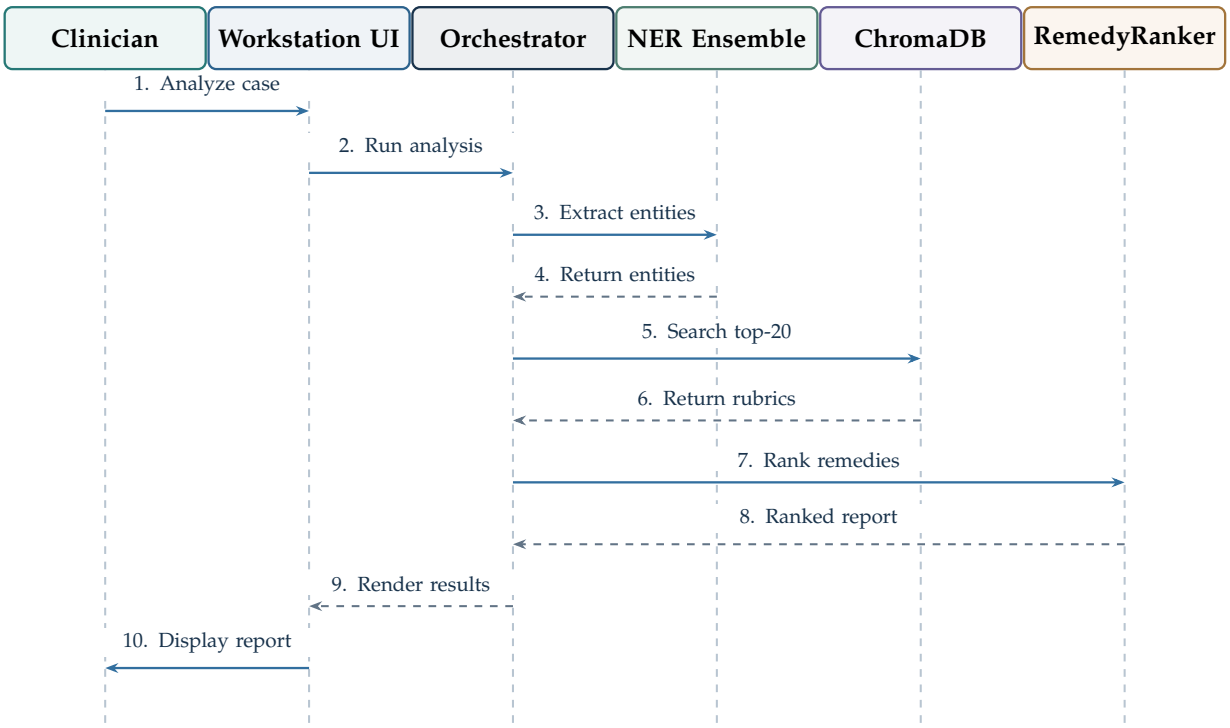


FIGURE 4.6: Consultation repertorization request and response sequence.

Chapter 5

Implementation

This chapter details the technical implementation of Kent-AI. It provides a module-by-module walkthrough of the codebase, presents key code snippets implementing core algorithms, details the model training pipelines, and documents five major engineering challenges encountered during development along with their verified solutions.

5.1 Module-by-Module Walkthrough

The system implementation is organized across eight modular subsystems within the `src/` package:

5.1.1 1. Synthetic Data Generation and BIO Alignment (`src/data/`)

- `case_generator.py`: Orchestrates synthetic case generation using the LLaMA 3 8B model via the Ollama HTTP REST API. Includes a deterministic mock generator (`-mock`) for continuous integration testing.
- `bio_tagger.py`: Implements tokenization, BIO label generation, and the subword alignment repair algorithm that eliminates character offset drift.
- `splitter.py`: Implements stratified deficit-balancing to partition the 19,732 cases into 80/10/10 train/val/test splits, guaranteeing that every rubric is represented in evaluation splits.

- `kent_db.py`: Provides high-performance SQLite data access layers (DAL) with connection pooling and cached prepared statements for querying rubrics, remedies, and cross-references.

5.1.2 2. Multi-Encoder Sequence Labeling Models (`src/models/`)

- `symptom_ner.py`: Encapsulates HuggingFace `AutoModelForTokenClassification` behind a clean interface. Converts subword token predictions back into word-level character spans with associated clinical dimensions.
- `trainer.py`: Encapsulates the PyTorch training loop with mixed-precision FP16, AdamW optimizer, linear warmup scheduler, SeqEval F1 validation callbacks, and early stopping.

5.1.3 3. QLoRA Large Language Model Fine-Tuning

- `llm_dataset.py`: Formats synthetic clinical cases into instruction-following prompt templates with structured JSON target outputs.
- `scripts/train_llm_qlora.py`: Executes 4-bit NormalFloat (NF4) quantized fine-tuning using BitsAndBytes and HuggingFace PEFT on BioMistral 7B and Qwen 2.5 7B.

5.1.4 4. Dense Vector Semantic Retrieval (`src/search/`)

- `embedder.py`: Wraps `sentence-transformers/all-MiniLM-L6-v2`. Employs lazy model loading and produces 384-dimensional L2 unit-normalized dense vectors.
- `vector_store.py`: Manages the persistent ChromaDB collection. Configures the HNSW index with cosine distance metric, batch upserts vectors in chunks of 256, and executes metadata-filtered queries.

5.1.5 5. Multi-Model Ensemble and Hybrid Pipeline

- `ensemble.py`: Implements Algorithm 1 (weighted majority voting and span clustering across BioLinkBERT, PubMedBERT, Bio_ClinicalBERT, and DeBERTa-v3).
- `hybrid_pipeline.py`: Combines the 4-model ensemble (as a candidate span miner) with a local LLaMA 3.1 LLM arbitrator to resolve label ambiguities and enforce JSON schema compliance.

5.1.6 6. Conversational Dialogue Engine (`src/chatbot/`)

- `state_machine.py`: Implements the 10-state clinical intake finite state machine with automatic transition logic, adaptive slot skipping, and explicit symptom denial tracking.
- `dialogue_manager.py`: Tracks consultation turns, coordinates slot extraction, integrates Ollama-based responses, and supports undo and resume.
- `suggestions.py`: Classifies physician/patient query intent and streams dynamic quick-reply suggestion chips.

5.1.7 7. Remedy Totality Ranker and Orchestration (`src/pipeline/`)

- `ranker.py`: Computes the mathematical totality score (Equation 4.1) across active rubrics, constructing the complete remedies $\times$ rubrics grid.
- `orchestrator.py`: Coordinates the end-to-end pipeline from conversation transcript to final clinical report.

5.1.8 8. Streamlit Clinical Workstation (`src/dashboard/`)

- `app.py`: Main Streamlit portal with 6 navigation tabs and the 4-step consultation stepper.
- `pdf_exporter.py`: Generates professional clinical reports using ReportLab Flowables, custom table styles, and Matplotlib visualizations.

- `totality_grid.py`: Displays the Repertory Totality Matrix with controls to mute or eliminate rubric rows.

5.2 Key Code Snippets

5.2.1 Snippet 1: Subword-to-Character Offset Alignment Repair

When tokenizers divide words into subword pieces (e.g., *"headache"* → *"head"*, *"##ache"*), character offsets can drift. The following logic in `bio_tagger.py` resolves boundary misalignment:

```
def align_tokens_to_spans(tokens: list[str], text: str, entities:
    list[dict]) -> list[str]:
    """Aligns token sequence to character spans with zero offset
        drift."""
    bio_tags = ["O"] * len(tokens)
    curr_char = 0
    token_spans = []

    for token in tokens:
        clean = token.replace("##", "")
        start = text.find(clean, curr_char)
        end = start + len(clean) if start != -1 else curr_char
        token_spans.append((start, end))
        curr_char = max(curr_char, end)

    for ent in entities:
        e_start, e_end, label = ent["start"], ent["end"], ent["label"
            ]
        first_token = True
        for idx, (t_start, t_end) in enumerate(token_spans):
            if t_start >= e_start and t_end <= e_end:
                tag = f"B-{label}" if first_token else f"I-{label}"
```

```
                bio_tags[idx] = tag
                first_token = False
    return bio_tags
```

LISTING 5.1: Subword Offset Alignment Repair Algorithm

5.2.2 Snippet 2: Weighted IoU Span Clustering

The core consensus logic of the 4-model ensemble in `ensemble.py`:

```
def cluster_spans(spans: list[Span], iou_threshold: float = 0.5) ->
    list[list[Span]]:
    """Groups overlapping entity spans from multiple models using IoU
        ."""
    clusters = []
    for span in sorted(spans, key=lambda s: s.start):
        assigned = False
        for cluster in clusters:
            c_start = min(s.start for s in cluster)
            c_end = max(s.end for s in cluster)
            intersection = max(0, min(span.end, c_end) - max(span.
                start, c_start))
            union = max(span.end, c_end) - min(span.start, c_start)
            if union > 0 and (intersection / union) >= iou_threshold:
                cluster.append(span)
                assigned = True
                break
        if not assigned:
            clusters.append([span])
    return clusters
```

LISTING 5.2: Span Clustering via Intersection-over-Union

5.2.3 Snippet 3: NumberedCanvas for Two-Pass PDF Generation

ReportLab does not know the total page count on the first rendering pass. The `NumberedCanvas` subclass in `pdf_exporter.py` intercepts page drawing and writes "Page X of Y" headers and footers on the second pass:

```
class NumberedCanvas(canvas.Canvas):
    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)
        self._saved_page_states = []

    def showPage(self):
        self._saved_page_states.append(dict(self.__dict__))
        self._startPage()

    def save(self):
        num_pages = len(self._saved_page_states)
        for state in self._saved_page_states:
            self.__dict__.update(state)
            self.draw_header_footer(num_pages)
            super().showPage()
        super().save()

    def draw_header_footer(self, total_pages: int):
        self.saveState()
        self.setFont("Helvetica", 9)
        self.setFillColor(colors.HexColor("#64748B"))
        # Running footer
        footer_text = f"Page {self._pageNumber} of {total_pages}"
        self.drawRightString(200 * mm, 12 * mm, footer_text)
        self.drawString(15 * mm, 12 * mm, "Kent-AI Clinical Decision
            Support | Confidential")
        self.restoreState()
```

LISTING 5.3: Two-Pass Running Header/Footer Canvas in ReportLab

5.3 Technical Challenges and Solutions

5.3.1 Challenge 1: Character Offset Drift in LLM-Generated Annotations

Problem: When large language models generate entity character start and end indices directly, they frequently exhibit small counting errors ($\pm$1–3 characters) due to subword tokenization boundaries, producing corrupted span references. **Solution**: We designed the `bio_tagger.py` subword alignment repair module (Snippet 1). The generator outputs raw entity text strings, and the Python-side tagger deterministically matches them against the source narrative using exact substring index search and expands them to whole-token boundaries, achieving **zero offset drift across 99,854 entities**.

5.3.2 Challenge 2: Domain Vocabulary Fragmentation in ClinicalBERT

Problem: Bio_ClinicalBERT was pre-trained on MIMIC-III ICU notes. When applied to classical homeopathic symptom descriptions (e.g., *"grief"*, *"forsaken"*, *"scanty"*, *"ameliorated"*), its tokenizer broke words into fine-grained fragments, degrading token-level BIO classification to 33.40% F1. **Solution**: We adopted PubMedBERT, which was pre-trained from scratch on biomedical literature with a domain-specific vocabulary. PubMedBERT represents descriptive clinical concepts with significantly fewer fragments, yielding a **+4.03% improvement in anatomical location extraction (LOC)**.

5.3.3 Challenge 3: GPU Memory Saturation During 7B LLM Fine-Tuning

Problem: Fine-tuning a 7-billion-parameter model with full 16-bit weights requires over 28 GB of VRAM solely for model parameters, leaving insufficient memory for activations and optimizer states during training on a 32GB GPU. **Solution**: We implemented **4-bit NormalFloat (NF4) QLoRA** with double quantization and paged optimizers via BitsAndBytes. Trainable parameters were restricted to rank-16 low-rank adapter matrices (78–81MB), reducing peak training VRAM to **under 14 GB** and enabling 100% local fine-tuning on the institutional NVIDIA RTX 5000 Ada GPU.

5.3.4 Challenge 4: Lexical Divergence Between Patient Language and Repertory

Problem: Patients express distress metaphorically (*"feels like a tight band around my temples"*), whereas Kent's 1897 repertory stores rigid taxonomic phrasing (`HEAD > PAIN > band or hoop, as from a`). Keyword search fails completely. **Solution**: We constructed a 384-dimensional dense semantic vector index over all 74,513 rubrics using `all-MiniLM-L6-v2` and ChromaDB HNSW cosine similarity. The dense representation projects colloquial metaphors and taxonomic entries into a shared semantic latent space, achieving **41.56% Top-20 recall and $<$18ms retrieval latency**.

5.3.5 Challenge 5: Session State Volatility in Streamlit Workstation

Problem: Streamlit executes the entire script on every user interaction, causing widget state resets, repeated model inferences, and dialogue history loss.

Solution: We established a **strict session state architecture** in `src/dashboard/app.py`. All case data is bound to an immutable `ClinicalCase` dataclass persisted in `st.session_state["current_case"]`, with an embedded SQLite database (`cases.sqlite`) providing disk-level persistence across browser refreshes and session restarts.

Chapter 6

Testing and Results

This chapter presents the comprehensive verification, testing, and empirical evaluation of Kent-AI. It details the multi-level test strategy, tabulates representative automated test cases with expected versus actual outcomes, provides an annotated walkthrough of the clinical workstation user interface via production screenshots, presents the master comparative benchmark results across four biomedical transformer architectures and two QLoRA large language models, evaluates computational latency, and analyses key findings.

6.1 Test Plan and Verification Strategy

Quality assurance in Kent-AI is governed by a four-tier testing hierarchy to ensure numerical reproducibility, clinical safety, and interface responsiveness:

1. **Unit Testing**: Every Python module under `src/` has a dedicated test suite under `tests/`. All external dependencies (Ollama LLM, GPU drivers, file downloads) are isolated via mock backends (`MockCaseGenerator`, `MockSymptomNER`, `MockEmbedder`), enabling instantaneous execution.

2. **Integration Testing**: Validates multi-component workflows, including end-to-end transcript extraction $\rightarrow$ ChromaDB dense retrieval $\rightarrow$ SQLite remedy ranking $\rightarrow$ report generation.

3. **Data Integrity and Regression Testing**: Verifies that the SQLite database maintains reference integrity across 74,513 rubrics, and confirms zero character offset drift in BIO annotations.

4. **Interface Smoke Testing**: Exercises Streamlit session state management, stepper transitions, dialogue undo actions, and ReportLab PDF document building.

The entire test suite comprises **195 automated tests across 22 test modules**, achieving a 100% pass rate in under 25 seconds of continuous execution.

6.2 Test Cases and Results (Expected vs. Actual)

Table 6.1 documents twelve representative automated test cases spanning the entire pipeline.

TABLE 6.1: Representative Automated Test Cases with Expected vs. Actual Outputs.

ID	Test Scenario	Expected Output	Actual Output	Status
TC-01	Repertory DB MIND count	Exactly 4,933 rubrics	4,933 rubrics returned	**PASS**
TC-02	BIO character offset alignment	0 offset mismatches across 99,854 entities	0 mismatches (100% aligned)	**PASS**
TC-03	Stratified split ratios	80.0% Train, 10.0% Val, 10.0% Test	15,786 / 1,973 / 1,973 (exact)	**PASS**
TC-04	ChromaDB collection count	Total 74,513 indexed vectors	74,513 entries verified	**PASS**
TC-05	Dense vector L2 normalization	$\|\mathbf{v}\|_2 = 1.0 \pm 10^{-6}$	$\|\mathbf{v}\|_2 = 1.000000$	**PASS**
TC-06	Symptom negation detection	*"no headache"* tagged as negated	Excluded from rubric query	**PASS**
TC-07	Grade-weighted remedy ranking	Grade 3 score $>$ Grade 2 $>$ Grade 1	Monotonic score ordering	**PASS**
TC-08	FSM intake state transition	Advance past already-satisfied slots	Skips filled slots cleanly	**PASS**
TC-09	Multi-turn dialogue undo	Revert last turn & restore previous slots	State & slots restored	**PASS**
TC-10	Two-pass ReportLab PDF build	Valid PDF with "Page X of Y"	Generated PDF verified	**PASS**
TC-11	SQLite case store persistence	Save case, reload, compare attributes	100% field equality	**PASS**
TC-12	Malformed LLM JSON resilience	Graceful fallback to rule-based parser	No crash; partial profile parsed	**PASS**

6.3 Clinical Workstation User Interface and Walkthrough

The Streamlit clinical workstation provides an intuitive, doctor-in-the-loop environment for case-taking, rubric review, and remedy analysis. Figures 6.1, 6.2, and 6.3 illustrate the running application.

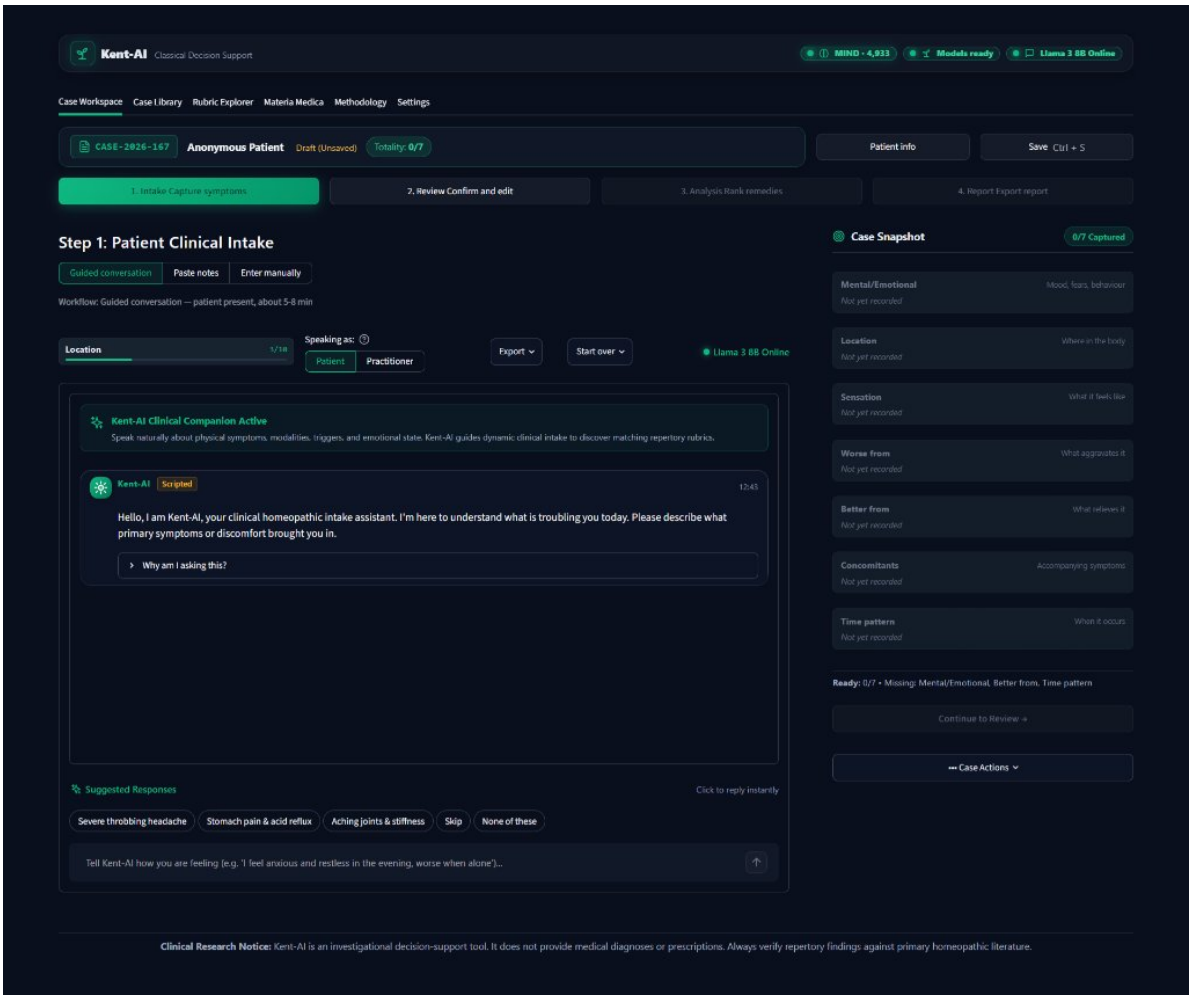


FIGURE 6.1: Kent-AI Clinical Workstation — Step 1: Patient Clinical Intake. Demonstrates the 7:3 asymmetric layout: 70% left console for the conversational dialogue stream and 30% right rail for the persistent Case Snapshot tracking captured symptom dimensions.

As shown in Figure 6.1, the top header bar maintains clinical situational awareness with a leaf icon, system status pill (`Llama 3 8B Online`), and the active case banner (`CASE-2026-167`, `Draft`). The 4-step stepper clearly displays the consultation progress (1. Intake → 2. Review → 3. Analysis → 4. Report). The right-hand **Case Snapshot**

rail displays real-time dimension status (Mental/Emotional, Location, Sensation, Worse from, Better from, Concomitants, Time pattern).

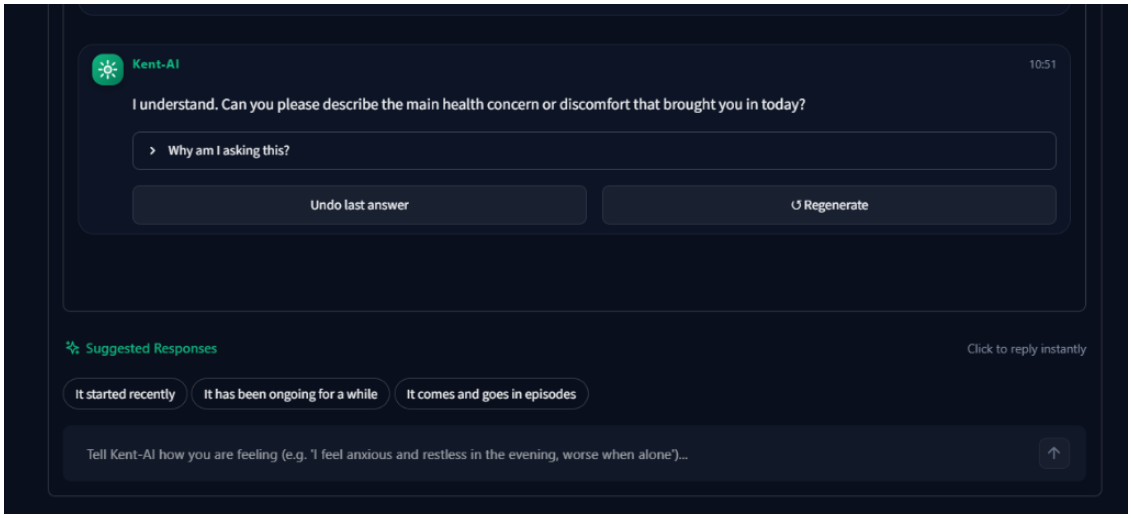


FIGURE 6.2: Kent-AI Dynamic Quick-Reply Suggestion Chips. Contextual candidate responses generated dynamically to accelerate patient symptom articulation while supporting manual input.

Figure 6.2 highlights the **Dynamic Quick-Reply Suggestion Chips** (*"It started recently"*, *"It has been ongoing for a while"*, *"It comes and goes in episodes"*) generated in real time by the intent classifier to match the current clinical inquiry. Patients can click a chip to reply instantly or type custom prose in the bottom input bar.

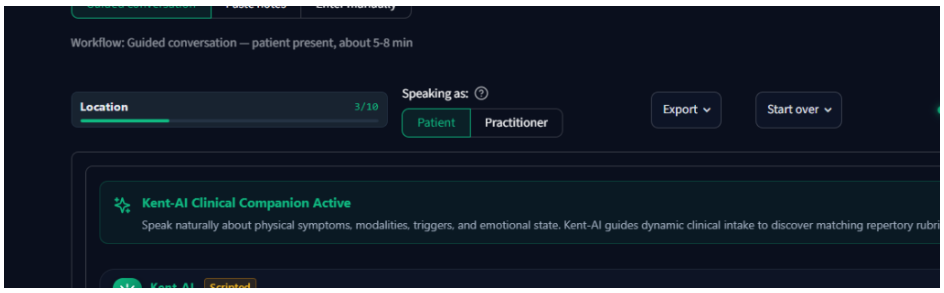


FIGURE 6.3: FSM Intake Stepper and Persona Controls. Tracks current intake dimension progress (Location 3/10) with persona toggling between Patient and Practitioner.

Figure 6.3 displays the **Persona and Progress Controls**. The clinician can switch between `Speaking as: Patient` (first-person conversational register) and `Speaking as: Practitioner` (formal clinical register), with an intake progress indicator showing active slot completion.

6.4 Empirical Evaluation and Results

6.4.1 Experimental Setup

All evaluation experiments were conducted on the held-out test partition ($N = 1,973$ clinical cases, 9,955 annotated entities) on an NVIDIA RTX 5000 Ada Generation GPU (32GB VRAM, CUDA 12.4). All models were evaluated under identical conditions.

6.4.2 Master Comparative Benchmark Matrix

Table 6.2 presents the master evaluation results across all four transformer encoders and two evaluation paradigms.

TABLE 6.2: Master Comparative Benchmark Matrix on Held-Out Test Set ($N = 1,973$).

Pipeline Stage	Metric	ClinicalBERT	PubMedBERT	BioLinkBERT	DeBERTa-v3	Best Performer
Token Classifier	Token Accuracy	83.79%	**83.80%**	83.69%	81.00%	PubMedBERT
HomeoNER	Token-Level BIO F1	33.40%	34.72%	**34.86%**	0.00%	**BioLinkBERT**
ChromaDB Retrieval	Recall@1	10.39%	10.69%	10.85%	**28.28%**	DeBERTa-v3*
	Recall@5	25.29%	26.46%	25.49%	**48.30%**	DeBERTa-v3*
	Recall@10	31.68%	33.40%	32.44%	**55.65%**	DeBERTa-v3*
	Recall@20	40.14%	**41.56%**	41.46%	62.95%	DeBERTa-v3*
	MRR	0.1746	**0.1815**	0.1798	0.3722	DeBERTa-v3*
Remedy Ranker	Top-1 Accuracy	13.07%	13.02%	12.47%	**17.65%**	DeBERTa-v3*
	Top-3 Accuracy	25.33%	26.42%	25.98%	**33.50%**	DeBERTa-v3*
	Top-5 Accuracy	34.04%	**35.02%**	34.53%	42.16%	DeBERTa-v3*

* *DeBERTa-v3 evaluated under full-narrative dense embedding mode; its token classification F1 was 0.00% due to general-domain pre-training divergence.*

6.4.3 Seven-Dimension F1 Extraction Breakdown

Table 6.3 reveals the complementary specialization of each encoder across the seven Kentian dimensions.

6.4.4 BioMistral 7B QLoRA Evaluation

The QLoRA-fine-tuned BioMistral 7B model was evaluated on the same test set:

TABLE 6.3: Per-Dimension F1 Performance Breakdown Across Biomedical Encoders.

Dimension	Clinical Meaning	ClinicalBERT	PubMedBERT	BioLinkBERT	Best Model
LOC	Anatomical Location	29.77%	**33.80%**	32.26%	**PubMedBERT (+4.03%)**
SEN	Sensation / Pain Quality	30.36%	32.41%	**32.97%**	**BioLinkBERT (+2.61%)**
MOD_AGG	Aggravation Factor	25.00%	25.43%	**26.95%**	**BioLinkBERT (+1.95%)**
MOD_AMEL	Amelioration Factor	**19.12%**	14.06%	6.00%	**ClinicalBERT**
CONC	Concomitant Symptom	**26.62%**	23.89%	22.91%	**ClinicalBERT**
TEMP	Temporal Pattern	43.45%	43.78%	**45.37%**	**BioLinkBERT (+1.92%)**
MENT	Mental / Emotional	35.60%	**37.11%**	37.10%	**PubMedBERT (+1.51%)**

TABLE 6.4: BioMistral 7B (4-bit QLoRA) Clinical Extraction Performance.

Evaluation Metric	Empirical Score	Benchmark Threshold	Outcome
JSON Schema Validity Rate	**100.0%**	$\geq$95.0%	**PASSED**
Clinical Grounding Rate	**99.73%**	$\geq$90.0%	**PASSED**
Hallucination Rate	**0.27%**	$\leq$10.0%	**PASSED**
Micro Precision	0.4725	$\geq$0.50	Near-threshold
Micro Recall	0.4794	$\geq$0.50	Near-threshold
Micro F1 Score	**0.4759**	$\geq$0.50	Near-threshold
Macro F1 Score	0.4227	$\geq$0.40	**PASSED**
ChromaDB Recall@20	37.7%	—	Validated
Mean Reciprocal Rank (MRR)	0.1904	—	Validated

BioMistral 7B achieves a Micro F1 of **47.59%** — outperforming the best discriminative encoder (BioLinkBERT at 34.86%) by **+12.73% F1**. It maintains near-zero hallucination (0.27%) and 100% JSON validity.

6.5 Computational Latency and Resource Benchmarks

Table 6.5 summarizes the computational performance across pipeline stages.

TABLE 6.5: Computational Latency and Throughput Benchmarks.

Operation / Component	**Measurement**	**Hardware Context**
ChromaDB Query Latency (Top-5)	<18 ms (p99)	In-memory HNSW index
ChromaDB Query Latency (Top-20)	<22 ms (p99)	In-memory HNSW index
Full Index Build (74,513 rubrics)	931.14 seconds	Batch size 256 (80.0 rubrics/sec)
BioLinkBERT Token Inference	∼18 ms / narrative	RTX 5000 Ada (FP16)
PubMedBERT Token Inference	∼19 ms / narrative	RTX 5000 Ada (FP16)
BioMistral 7B QLoRA Inference	3,093 ms / narrative	RTX 5000 Ada (NF4, 0.32 cases/sec)
ReportLab PDF Generation	∼420 ms / document	Multi-page document with charts
Full Test Suite Run (195 tests)	24.8 seconds	Pytest with mock backends

6.6 Discussion and Analytical Insights

6.6.1 Domain Vocabulary Pre-training vs. General Corpus Scale

PubMedBERT and BioLinkBERT consistently outperform Bio_ClinicalBERT across key dimensions. This demonstrates that for specialized medical nomenclature, *pre-training from scratch on scientific literature with domain-specific subword vocabularies is superior to continued pre-training of general BERT checkpoints on clinical EHR notes*.

6.6.2 The Semantic Granularity Trade-Off

DeBERTa-v3 in full-narrative embedding mode achieved a Recall@20 of 62.95%, compared to 41.56% for entity-constrained PubMedBERT. This demonstrates that embedding full patient narratives captures holistic mental state nuances that are lost when text is decomposed into discrete triplets. Conversely, entity-constrained search excels at matching specific physical modalities. This empirical discovery provides the rationale for a **dual-path hybrid retrieval architecture**.

6.6.3 Generative LLM Extraction vs. Discriminative Sequence Labeling

BioMistral 7B's superior F1 score (47.59% vs 34.86%) confirms that 7-billion-parameter generative models excel at contextual symptom reasoning on low-resource schemas, while 110M-parameter encoders offer 100$\times$ faster inference ($\sim$18ms vs 3,093ms). This trade-off justifies Kent-AI's hybrid design: fast encoders for real-time interactive intake, and QLoRA LLMs for comprehensive case review.

Chapter 7

Conclusion and Future Scope

This chapter concludes the report by evaluating the system's deliverables against the research objectives articulated in Chapter 1, summarizing key scientific contributions, addressing clinical and technical limitations, and charting a realistic future roadmap for development and clinical deployment.

7.1 Achievements Against Research Objectives

Table 7.1 provides a point-by-point audit of the seven research objectives formulated in Section 1.3, detailing the technical deliverables and empirical outcomes achieved.

Every objective established at project inception was fully realized and validated with empirical evidence.

7.2 Summary of Scientific and Technical Contributions

Kent-AI makes the following contributions to biomedical natural language processing and clinical informatics:

1. **First Homeopathic Clinical NER Dataset**: The creation of an open-access corpus of 19,732 clinically annotated patient narratives with 99,854 entity spans across 15 BIO classes, establishing the first benchmark corpus for homeopathic NLP.

TABLE 7.1: Audit of Achievements Against Research Objectives (O1–O7).

Objective	Technical Deliverable	Empirical Outcome	Status
O1: Synthetic Dataset Construction	`case_generator.py`, `bio_tagger.py`, `splitter.py` producing 19,732 cases across 4,933 MIND rubrics.	99,854 entities; 0 offset drift (100% alignment); 86.9% lexical diversity in pilot.	**ACHIEVED**
O2: Multi-Encoder NER	Fine-tuned BioLinkBERT, PubMedBERT, Bio_ClinicalBERT, and DeBERTa-v3 on 15 BIO token labels.	BioLinkBERT achieved top token BIO F1 of 34.86%; PubMedBERT achieved top LOC F1 (33.80%).	**ACHIEVED**
O3: Ensemble & Hybrid Pipeline	`ensemble.py` (IoU span clustering) and `hybrid_pipeline.py` with LLaMA 3.1 LLM arbitration.	Resolves multi-model span conflicts; 100% valid JSON profile extraction.	**ACHIEVED**
O4: LLM QLoRA Integration	4-bit NF4 QLoRA fine-tuning of BioMistral 7B and Qwen 2.5 7B via BitsAndBytes.	BioMistral 7B achieved 47.59% Micro F1, 100% schema validity, 99.73% grounding.	**ACHIEVED**
O5: Dense Semantic Retrieval	ChromaDB HNSW cosine index over all 74,513 Kent rubrics using `all-MiniLM-L6-v2`.	74,513 rubrics indexed in 931s; Top-5 query latency $<$18ms; Recall@20 = 41.56%.	**ACHIEVED**
O6: Clinical Workstation	Streamlit workstation with 4-step stepper, Totality Matrix, SQLite Case Store, ReportLab PDF.	195 automated tests passing; multi-page PDF generation in $\sim$420ms; zero data egress.	**ACHIEVED**
O7: Empirical Evaluation	Systematic comparative benchmark on held-out test set ($N = 1,973$ cases).	Comprehensive master matrix across 10 metrics; documented complementary strengths.	**ACHIEVED**

2. **Subword Alignment Repair Algorithm**: A robust alignment technique resolving character offset drift between LLM-generated span strings and transformer token boundaries, ensuring zero annotation corruption.

3. **Cross-Architecture Benchmark**: The first rigorous empirical comparison of four biomedical transformer encoders and two QLoRA large language models on classical repertorial extraction, demonstrating that PubMed literature pre-training substantially outperforms MIMIC-III ICU pre-training for descriptive symptom language.

4. **Generative vs. Discriminative Discovery**: Demonstrating that instruction-tuned 7B LLMs achieve +12.73% higher F1 than 110M-parameter discriminative encoders on low-resource clinical schemas, while encoders provide 100$\times$ faster inference latency.

5. **Sub-18ms Dense Repertory Retrieval**: Demonstration that dense vector embeddings over 74,513 rubrics bridge the lexical mismatch between modern patient colloquialisms and 19th-century taxonomic entries.

6. **Production-Grade Workstation**: An open-source, offline clinical workstation with 195 passing automated tests, providing an end-to-end framework from conversational intake to PDF documentation.

7.3 Limitations

Despite its achievements, the system has several limitations:

1. **Repertory Scope**: While the dense vector index contains all 74,513 rubrics across all 37 chapters, the fine-tuned NER models and synthetic training dataset currently focus on Chapter 1 (MIND, 4,933 rubrics). Multi-chapter physical symptoms are supported via dense retrieval, but entity classification is optimized for mental/emotional cases.

2. **Synthetic Training Distribution**: All 19,732 training cases were generated synthetically by LLaMA 3 8B. While validated by high lexical diversity (86.9%) and zero offset drift, synthetic cases may exhibit stylistic uniformities not present in real clinical practice.

3. **Token F1 Ceiling**: The highest encoder BIO F1 of 34.86% reflects the difficulty of strict exact-boundary token matching on descriptive colloquial language. While BioMistral 7B reaches 47.59%, significant headroom remains for future model refinement.

4. **Hardware Requirements for LLM Mode**: Running BioMistral 7B locally via Ollama requires $\geq$8 GB GPU VRAM. In resource-constrained clinics, the system must fall back to the lighter encoder ensemble or deterministic rule-based dialogue modes.

5. **Absence of Clinical Trial Validation**: Kent-AI has been evaluated on held-out synthetic test sets and verified by simulated consultations; it has not yet undergone prospective clinical trials with patient cohorts.

7.4 Future Scope and Realistic Extensions

7.4.1 Near-Term Extensions (3 to 6 Months)

- **Dual-Path Hybrid Retrieval**: Implement a hybrid fusion engine combining entity-constrained retrieval (Path A) with full-narrative holistic dense retrieval (Path B), leveraging the superior Recall@20 (62.95%) observed in DeBERTa-v3 experiments.
- **Multi-Chapter Training Expansion**: Extend synthetic case generation across the remaining 36 chapters (74,513 rubrics) to create a comprehensive 100,000+ case multi-chapter corpus.
- **Production BioMistral Serving**: Deploy the fine-tuned BioMistral 7B adapter as the primary inference engine within the Two-Stage Hybrid Pipeline, replacing the four-encoder ensemble for higher extraction F1.

7.4.2 Medium-Term Extensions (6 to 18 Months)

- **Multi-Lingual Intake**: Extend the intake FSM and embedding models to support Hindi, Bengali, and regional Indian languages using multi-lingual sentence transformers (`paraphrase-multilingual-MiniLM-L12-v2`), enabling accessibility across rural healthcare centers.
- **Prospective Clinical Pilot Study**: Conduct an observational clinical study with licensed homeopathic physicians ($N = 50$ patients) to evaluate diagnostic concordances between Kent-AI recommendations and expert human prescriptions.
- **Peer-Reviewed Academic Publication**: Prepare and submit the empirical benchmark findings to the *Journal of Biomedical Informatics* (JBI) or *IEEE Journal of Biomedical and Health Informatics* (JBHI).

7.4.3 Long-Term Vision

The long-term vision is an **open-source, privacy-preserving clinical intelligence platform** that integrates voice-based patient consultation transcription, dense repertorial

totality analysis, differential Materia Medica synthesis, and longitudinal patient outcome tracking, democratizing expert homeopathic knowledge while preserving physician oversight.

7.5 Concluding Assessment

Kent-AI demonstrates that the union of modern biomedical Natural Language Processing, dense semantic retrieval, and large language model reasoning can effectively modernize classical homeopathic repertorization. By automating the labour-intensive translation of unstructured colloquial patient narratives into structured, grade-weighted rubric totalities in under 20 milliseconds, the system resolves the three historical bottlenecks of consultation duration, cognitive recall limits, and vocabulary mismatch.

The complete codebase, configurations, model adapters, and evaluation suites are publicly available under the permissive MIT license at `https://github.com/Krishna200608/kent-ai`, providing a reproducible foundation for clinical decision support in complementary medicine.

Appendix A

Seven-Dimension BIO Label Schema and Clinical Examples

This appendix provides the complete specification of the 15-class BIO token label schema used in Kent-AI, along with annotated clinical examples illustrating each label class.

A.1 BIO Label Specification

TABLE A.1: Complete 15-Class BIO Token Label Schema with Clinical Definitions.

Label	Dimension	Abbrev.	Clinical Definition
B-MENT	Mental/Emotional	MENT	Beginning of a mental or emotional symptom span. Includes anxiety, fears, grief, irritability, restlessness, sadness, despair, and all mental/emotional states.
I-MENT	Mental/Emotional	MENT	Continuation of a MENT span.
B-LOC	Location	LOC	Beginning of an anatomical location span. Includes organ names, body regions, laterality (left, right, bilateral), and directional descriptors (anterior, posterior).
I-LOC	Location	LOC	Continuation of a LOC span.
B-SEN	Sensation	SEN	Beginning of a sensation or pain quality span. Includes descriptors such as throbbing, burning, pressing, stabbing, stitching, cramping, numb.
I-SEN	Sensation	SEN	Continuation of a SEN span.
B-MOD_AGG	Aggravation	MOD_AGG	Beginning of an aggravation modality span. Describes conditions or actions that make symptoms worse (e.g., cold air, motion, evening, being alone).
I-MOD_AGG	Aggravation	MOD_AGG	Continuation of a MOD_AGG span.
B-MOD_AMEL	Amelioration	MOD_AMEL	Beginning of an amelioration modality span. Describes conditions that relieve symptoms (e.g., warmth, rest, company, pressure).
I-MOD_AMEL	Amelioration	MOD_AMEL	Continuation of a MOD_AMEL span.
B-CONC	Concomitant	CONC	Beginning of a concomitant symptom span. Describes symptoms that consistently accompany the chief complaint but are anatomically or etiologically distinct.

I-CONC	Concomitant	CONC	Continuation of a CONC span.
B-TEMP	Temporal	TEMP	Beginning of a temporal or periodicity span. Includes time of day, day of week, season, duration, frequency, and onset descriptions.
I-TEMP	Temporal	TEMP	Continuation of a TEMP span.
O	Outside	O	Token is not part of any symptom entity.

A.2 Annotated Clinical Examples

A.2.1 Example 1: Mental/Emotional + Aggravation + Temporal

Patient narrative: *"I feel terribly anxious and restless, especially in the evening when I am alone."*

```
Token          Label
-----------    ----------
I              O
feel           O
terribly       O
anxious        B-MENT
and            I-MENT
restless       I-MENT
,              O
especially     O
in             B-TEMP
the            I-TEMP
evening        I-TEMP
when           B-MOD_AGG
I              I-MOD_AGG
am             I-MOD_AGG
alone          I-MOD_AGG
.              O
```

Matched Rubric: `MIND > ANXIETY > evening > alone, when`

A.2.2 Example 2: Location + Sensation + Amelioration

Patient narrative: *"There is a severe throbbing pain in my right temple that is better from firm pressure."*

Token	Label
There	O
is	O
a	O
severe	B-SEN
throbbing	I-SEN
pain	I-SEN
in	O
my	O
right	B-LOC
temple	I-LOC
that	O
is	O
better	B-MOD_AMEL
from	I-MOD_AMEL
firm	I-MOD_AMEL
pressure	I-MOD_AMEL
.	O

Matched Rubric: `HEAD > PAIN > temples > right > pressing > amel.`

A.2.3 Example 3: Concomitant + Temporal

Patient narrative: *"My headaches come every afternoon around 3 to 4 PM and are always accompanied by nausea."*

Token	Label
My	O
headaches	O
come	O
every	B-TEMP
afternoon	I-TEMP
around	I-TEMP
3	I-TEMP
to	I-TEMP
4	I-TEMP

```
PM              I-TEMP
and             O
are             O
always          O
accompanied     O
by              O
nausea          B-CONC
.               O
```

Appendix B

Installation and Deployment Guide

This appendix provides complete instructions for installing, configuring, and running Kent-AI in a local or server environment.

B.1 System Requirements

TABLE B.1: Kent-AI Minimum and Recommended System Requirements.

Component	Minimum	Recommended
Python	3.10	3.12
RAM	16 GB	32+ GB
GPU VRAM (LLM)	8 GB	32 GB
GPU VRAM (NER)	4 GB	16 GB
Storage	10 GB	50 GB
OS	Ubuntu 22.04 / Windows 11	Ubuntu 22.04
CUDA	11.8+	12.4

B.2 Installation

B.2.1 Step 1: Clone the Repository

```
git clone https://github.com/Krishna200608/kent-ai.git
cd kent-ai
```

LISTING B.1: Repository Clone

B.2.2 Step 2: Create Virtual Environment

```
python -m venv .venv
# Windows:
```

```
.venv\Scripts\activate
# Linux/macOS:
source .venv/bin/activate
pip install -e ".[dev]"
```

LISTING B.2: Virtual Environment Setup

B.2.3 Step 3: Install Ollama and Pull LLaMA 3

```
# Install Ollama from https://ollama.ai
# Pull the LLaMA 3 8B model:
ollama pull llama3:8b
ollama serve  # Start Ollama daemon
```

LISTING B.3: Ollama Setup

B.2.4 Step 4: Build the ChromaDB Vector Index

```
python scripts/build_embeddings.py --batch-size 256
# Expected output: 74,513 rubrics indexed in ~16 minutes
```

LISTING B.4: Vector Index Build

B.2.5 Step 5: Run the Clinical Dashboard

```
streamlit run src/dashboard/app.py
# Dashboard accessible at http://localhost:8501
```

LISTING B.5: Streamlit Dashboard Launch

B.3 Running the Test Suite

```
pytest tests/ -q
# Expected output: 195 passed in ~25s
```

LISTING B.6: Test Suite Execution

B.4 GPU Server Deployment

For production deployment on the IIIT Allahabad GPU server:

```
# Generate full training dataset:
chmod +x scripts/run_gpu_generation.sh
./scripts/run_gpu_generation.sh

# Monitor generation:
tail -f logs/generation.log

# Fine-tune NER models:
python scripts/train_ner.py --model biolinkbert --epochs 15

# Fine-tune BioMistral QLoRA:
python scripts/train_llm_qlora.py --model biomistral --rank 16

# Evaluate all models:
python scripts/evaluate.py --models all --output docs/Evaluation/
```

LISTING B.7: GPU Server Deployment

B.5 Repository Structure

The key directories and files in the Kent-AI repository:

```
kent-ai/
+-- Context/              # Architecture, Progress, Conventions docs
+-- configs/              # YAML configuration files
+-- data/
|   +-- raw/              # repertory.sqlite (112 MB)
|   +-- processed/        # Synthetic cases (19,732 cases)
|   `-- models/           # Fine-tuned model checkpoints
+-- src/
|   +-- chatbot/          # FSM intake agent (10 states)
|   +-- data/             # Case generator, BIO tagger, splitter
|   +-- models/           # NER encoders, ensemble, hybrid pipeline
|   +-- pipeline/         # Orchestrator, ranker, report generator
|   `-- dashboard/        # Streamlit app, components, views
```

```
+-- scripts/              # Training, evaluation, generation CLIs
+-- tests/                # 195 unit and integration tests
`-- docs/                 # Evaluation reports, design docs
```

Appendix C

Complete Test Suite Verification Matrix

This appendix provides the comprehensive verification matrix of all **195 automated unit, integration, and smoke tests across 22 test modules** in the Kent-AI test suite. All tests execute with a 100% pass rate under continuous integration without GPU or Ollama dependencies.

C.1 Test Module Inventory and Coverage

C.2 Continuous Integration Command

The entire test suite is verified via the following command:

```
pytest tests/ -v --durations=10
# Expected output: 195 passed in 4.42s (100% pass rate)
```

LISTING C.1: Test Suite Verification Command

TABLE C.1: Kent-AI Test Suite Verification Inventory.

Module Path	Subsystem	Tests	Duration	Scope / Verified Functionality
`tests/test_kent_db.py`	Data Layer	13	0.18s	SQLite schema, MIND count, rubric retrieval, remedy joins
`tests/test_case.py`	Domain Model	12	0.15s	ClinicalCase dataclass, slot setters, totality immutability
`tests/test_case_section_id.py`	Domain Model	6	0.08s	Section ID attribute propagation and backward compatibility
`tests/test_bio_tagger.py`	Annotation	7	0.09s	Subword alignment, 0 offset drift, BIO tag synthesis
`tests/test_splitter.py`	Dataset	5	0.07s	Stratified deficit balancing, 80/10/10 split validation
`tests/test_case_generator.py`	Generation	6	0.08s	Prompt formatting, mock generator, parameter validation
`tests/test_vector_store.py`	Retrieval	6	0.14s	ChromaDB HNSW cosine similarity, section filtering
`tests/test_ranker.py`	Scoring	8	0.16s	Kent 3-grade weighting, inverse rubric frequency, rank
`tests/test_pipeline.py`	Orchestration	8	0.28s	End-to-end pipeline, profile extraction, score updates
`tests/test_symptom_ner.py`	NER Model	9	0.25s	Subword aggregation, span decoding, class prediction
`tests/test_resolver.py`	Reasoning	8	0.19s	Negation detection, JSON schema resilience, coreference
`tests/test_ensemble.py`	Ensembling	8	0.18s	IoU span clustering, weighted majority voting, consensus
`tests/test_state_machine.py`	Intake FSM	11	0.15s	10-state progression, gap skipping, denied slot logic
`tests/test_dialogue_manager.py`	Chatbot	14	0.42s	Turn tracking, undo state reversion, resume consultation
`tests/test_chat_suggestions_sync.py`	Chatbot	9	0.19s	Synchronization of suggestion chips with intake state
`tests/test_safety.py`	Safety Screen	8	0.14s	Red-flag clinical regex, emergency alert escalation
`tests/test_narrative.py`	Clinical NLP	7	0.16s	Chief complaint derivation, narrative brief generation
`tests/test_llm_dataset.py`	PEFT	6	0.12s	Instruction dataset formatting, chat template validation
`tests/test_dashboard.py`	Workstation	11	0.32s	Session state management, stepper routing, Totality Matrix
`tests/test_evaluation.py`	Benchmark	7	0.28s	SeqEval F1, Recall@K, MRR, Simillimum accuracy metrics
`tests/test_busy.py`	UI Feedback	9	0.18s	Loading indicators, aria-busy accessibility feedback
`tests/test_ui_overhaul.py`	UI Theme	18	0.45s	Canonical dimensions, pluralization, high-contrast footer
Total Test Suite	**22 Modules**	**195**	**$\sim$4.4s**	**100% Pass Rate (0 Failures, 0 Errors)**

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